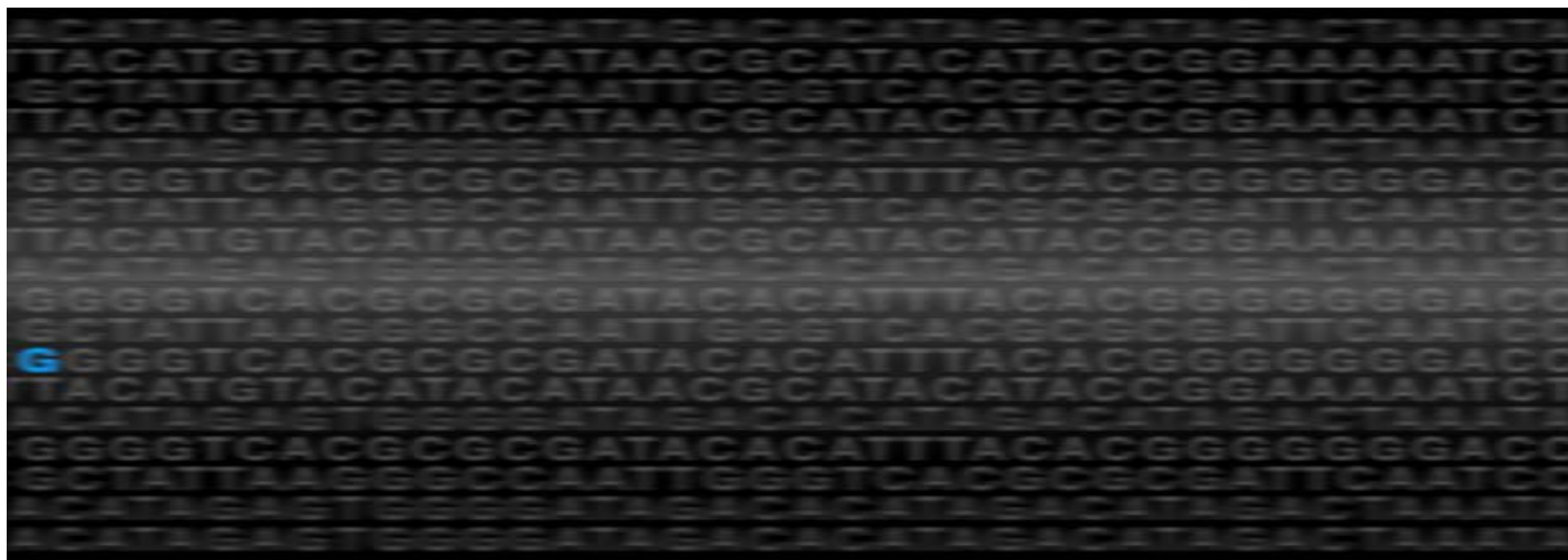


Conundrums and challenges of clinical variant interpretation



MAVE workshop
WTSI
Nov 2025

Clare Turnbull, PhD MEpi FFPH FRCP FRCPath

- **Professor in Cancer Genetics**, Institute of Cancer Research
- **NHS Consultant in Clinical Cancer Genetics** (Honorary), Royal Marsden NHS Foundation Trust
- **Consultant in Public Health Medicine** (Honorary), National Disease Registration Service, NHSE

Themes

1. Clinical Genetics overview
2. Phenotype and ascertainment
3. Variant-specific risks/genotype-phenotype
4. Evolution of Variant Interpretation Frameworks
5. MAVES, clinical variant interpretation and opportunities forward

Origins of clinical genetics: pre-'genomic era'

Specialists with knowledge of **patterns** of rare (inherited) disease and **emerging technologies**

- **Predictive testing:** Huntington
- **Prenatal testing:** Duchenne Muscular Dystrophy
- **Limited management opportunities:**

Making a diagnosis in children with ID/congenital abnormalities

- Dysmorphology:
 - Describing/grouping/splitting features
 - Syndromes linked to inheritance (AD, XR, XL plus teratogens, field defects etc)
- Lacked molecular basis (often still do)→ empirical predictions
- Forecasting (adult function), **recurrence risk**

Options for management in genomic era

- ie on detection in 'relevant Mendelian gene' of a germline pathogenic variant **ie a variant we assign as pathogenic**

Rare Paediatric Disease

Genetic Diagnosis

- **Diagnostic Odyssey** closure: SWAN
- **Phenotypic** Spectrum/monitoring for complications, eg renal, eyes
- **Drug (Trials):** enzyme replacement, gene therapy, mitigate disease progression

Reproductive Options



- **Pre-implantation Genetic Diagnosis**
- **Prenatal Testing:** amnio, CVS, cfDNA

Cancer Susceptibility Genes

Screening



Enhanced screening programmes:

- **Modality** **MRI**
- **Age of starting** **30**
- **Frequency** **annually**

Primary chemoprevention



- **RE-PURPOSED:** aspirin
- **RE-POSITIONED:** **tamoxifen**
- **TARGETED:** **Rank-Ligand inhibition, PARPi**

Behavioural



- **Sun exposure (CDKN2A)**
- **Hormonal factors**
- **Smoking**

Primary surgical prevention



- **Mastectomy (BRCA1/2)**
- **Oophorectomy (BRCA1/2)**
- **Colectomy (APC)**
- **Gastrectomy (CDH1)**

Complexities/challenges: Clinical Genetics Paradigms

- Diseases
 - very rare
 - Long-time course for emergence of symptoms
- Clinical management
 - heterogenous in space (local/national/international)
 - Technologies/costs change with time
 - Can not randomise
- → → Limited Evidence base:
 - clinical efficacy
 - cost effectiveness
- Evidence for efficacy of Clinical Management Interventions presumes dichotomy of gene penetrance/pathogenicity
 - ***Mutation present:*** complete (very high) penetrance for disease
 - ***Mutation absent:*** no (very low) risk of disease

Themes

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Phenotypes and ascertainment: Rare disease

What is phenotype?

Observable manifestations (of germline mutation).

- Gene → constellation of features
- 1) Significance of features
 - **Clinically Important** eg intellectual disability, invasive cancer
 - **Diagnostic “hard handles”** eg ‘dysmorphic’ facial features, skin changes
- 2) Ascertainment of features
 - **Overt:** apparent to all
 - Short stature
 - **Covert:** need to look for them (incl radiology, biochemistry)

What is phenotype?

“Covert” phenotypic features you only find if you look for them



Axillary freckling
(Neurofibromatosis I,
NF1)
Examine axilla



Ash leaf macules
(Tuberose Sclerosis,
TSC1/TSC2):
Woods lamp



CHRPE
(Familial Adenomatous
Polyposis, APC)
fundoscopy

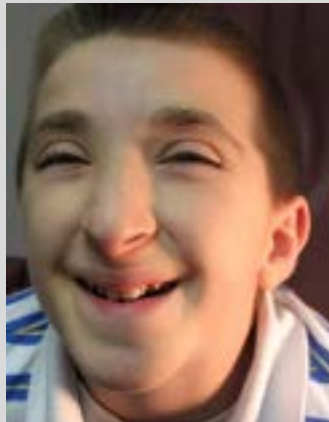
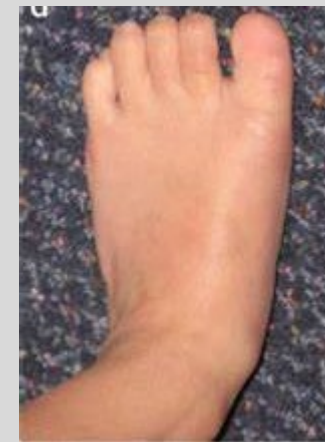
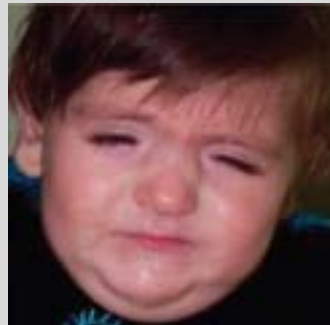


Jaw Cysts
(Hyperparathyroidism jaw
tumor CDC73):
Jaw film
(orthopantomography)

Rubinstein Taybi syndrome

CREBBP mutation

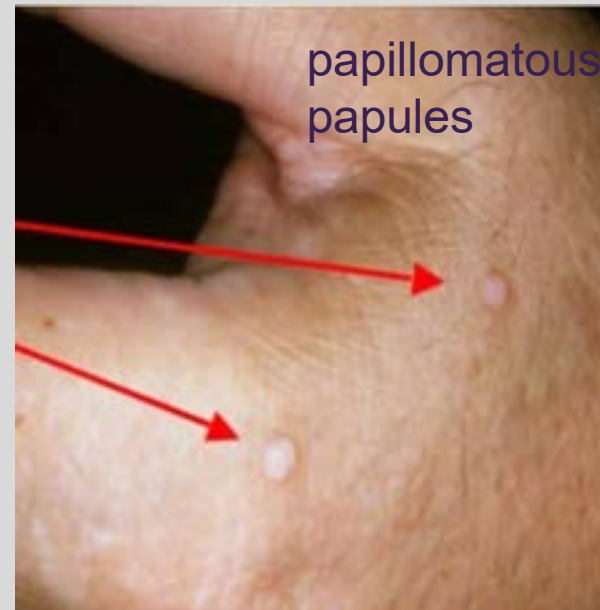
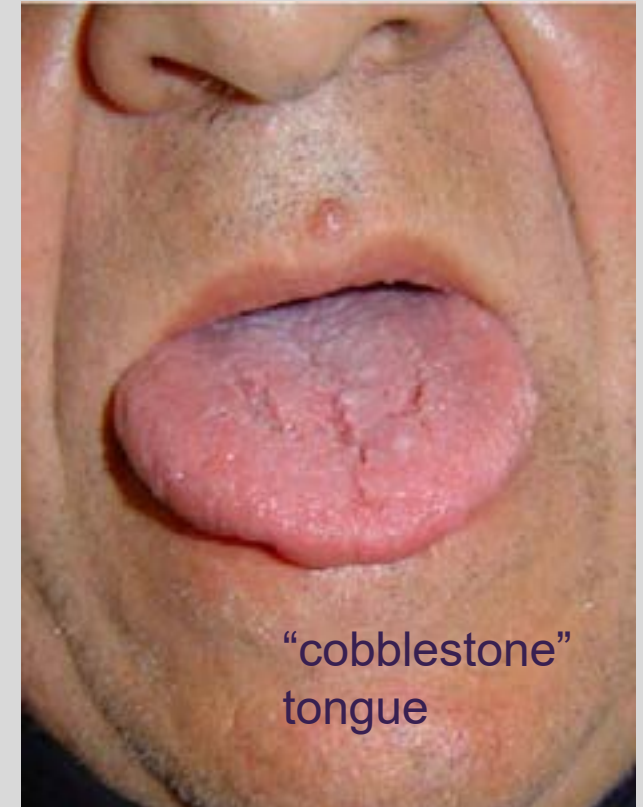
- **Moderate-severe intellectual disability**
- Short stature
- Distinctive facial features
 - down-slanting palpebral fissures
 - prominent beaked nose
- Broad thumbs and big toes



Cowden Syndrome

PTEN mutation

- Invasive cancers of thyroid, breast, endometrium, colorectum
- Harmatomas and other benign tumours
 - Skin:
 - Lipomas
 - papillomatous papules
 - Trichilemmomas
 - “cobblestone” tongue
 - freckles on their penis
 - acral and plantar keratoses
 - Benign fibrous breast disease
 - Thyroid hyperplasia
 - GI tract harmatomas
- Macrocephaly** (age/sex/ethnicity-specific curves; familial macrocephaly)





Phenotypes and ascertainment: Cancer susceptibility

Cancer Susceptibility Genes: Phenotypic spectrum

“associated with common cancers through rare syndromes”



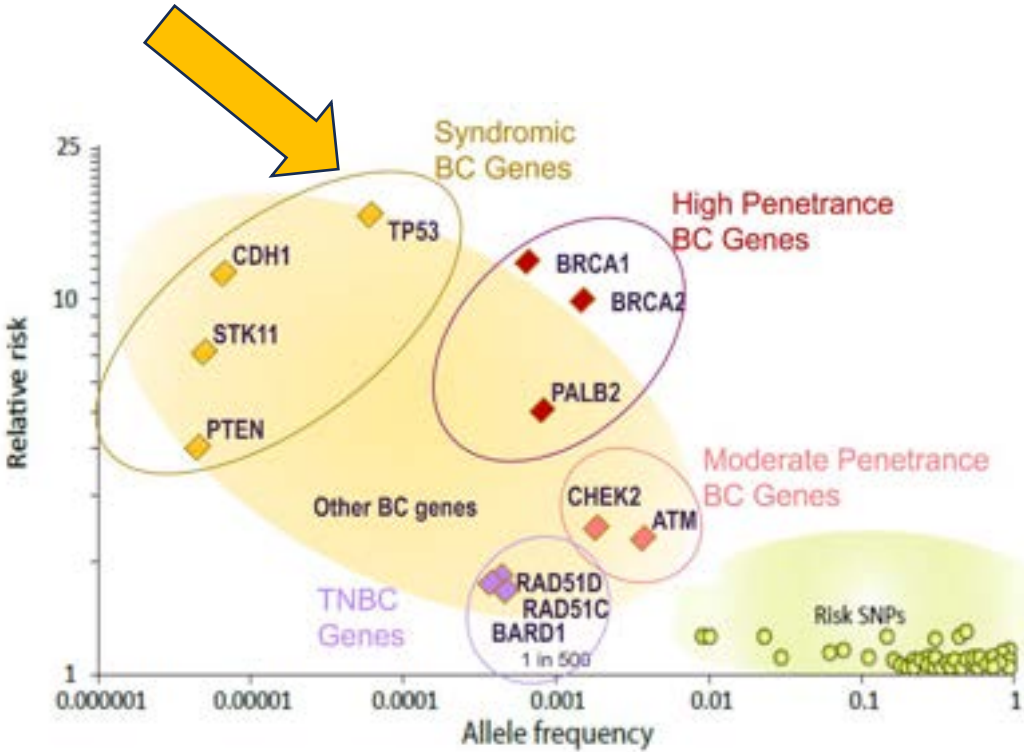


Phenotypes and ascertainment: Uncertainty regarding penetrance

Estimation of genomic risk (penetrance): familial (segregation) studies

	Provider	Total number of genes	BRCA1	BRCA2	PALB2	CHEK2	ATM	BARD1	RAD51C	RAD51D	TP53	PTEN	NF1	STK11	CDH1
Breast-cancer Specific Panel	UK NHSE NTD R208	7 (8)	✓	✓	✓	✓	✓	(✓)	✓	✓					
	Everything Genetics	12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
	Invitae	9	✓	✓	✓	✓	✓				✓	✓		✓	✓
	Oxford Gene Technology	7	✓	✓	✓	✓	✓				✓	✓			
	Centogene	27	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
	Check4Cancer	11	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	
	Fulgent Diagnostics	13	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Radbound UMC	10	✓	✓	✓	✓	✓	✓	✓	✓		✓			
	Catalan Servei de la Salut	15	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓
Breast-Ovarian panel	Randox	8	✓	✓	✓	✓	✓				✓	✓			✓
	Ambry Genetics	18	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓		✓
	GeneDx	24	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Illumina	11	✓	✓	✓	✓	✓		✓	✓	✓				✓
	Paragon Genomics	37	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓

Turnbull, et al. *Annals Onc* 2025, 36(8)



Estimation of genomic risk (penetrance): familial (segregation) studies

THE NEW ENGLAND JOURNAL of MEDICINE

SPECIAL REPORT

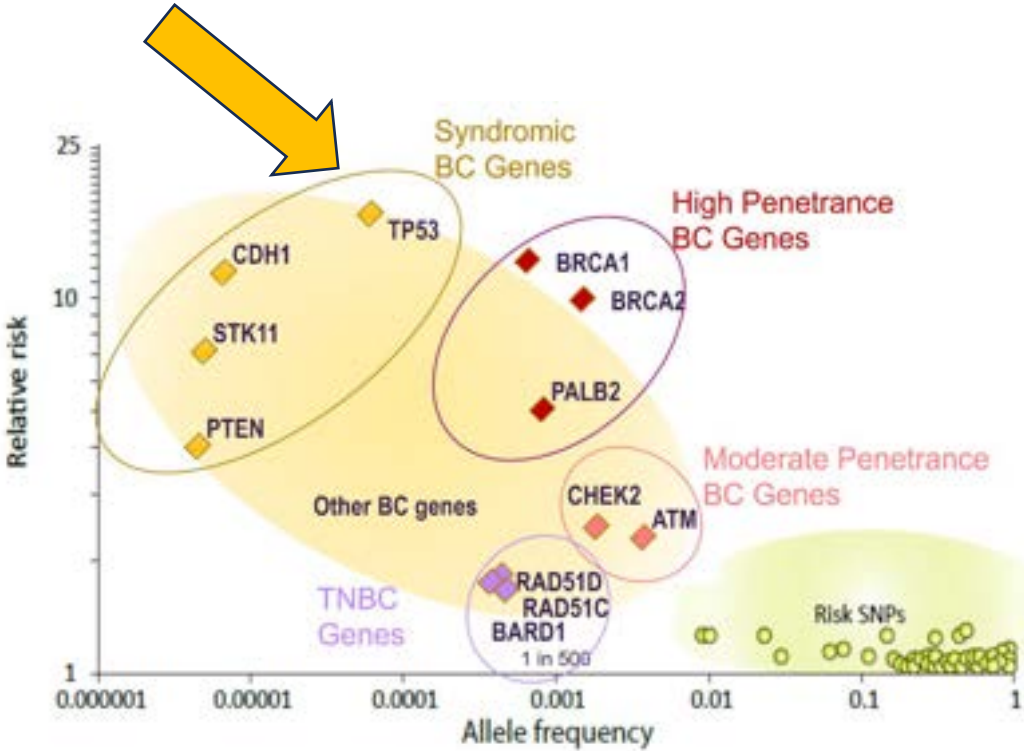
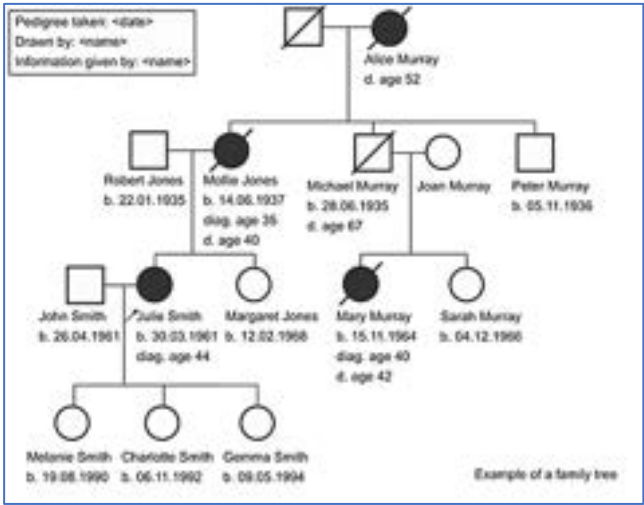
Gene-Panel Sequencing and the Prediction
of Breast-Cancer Risk

Douglas F. Easton, Ph.D., Paul D.P. Pharoah, Ph.D., Antonis C. Antoniou, Ph.D.,
Marc Tischkowitz, M.D., Ph.D., Sean V. Tavtigian, Ph.D., Katherine L. Nathanson, M.D.,
Peter Devilee, Ph.D., Alfons Meindl, Ph.D., Fergus J. Couch, Ph.D., Melissa Southey, Ph.D.,
David E. Goldgar, Ph.D., D. Gareth R. Evans, M.D., Georgia Chenevix-Trench, Ph.D.,
Nazneen Rahman, M.D., Ph.D., Mark Robson, M.D., Susan M. Domchek, M.D.,
and William D. Foulkes, M.B., B.S., Ph.D.

**Best estimate for TP53
breast cancer risk: OR=110**
(familial-based
ascertainment)

N ENGL J MED 372;23 NEJM.ORG JUNE 4, 2015

The New England Journal of Medicine



Turnbull C, Sud A, and Houlston, R.S. *Nat Genet.* 2018;50(9).

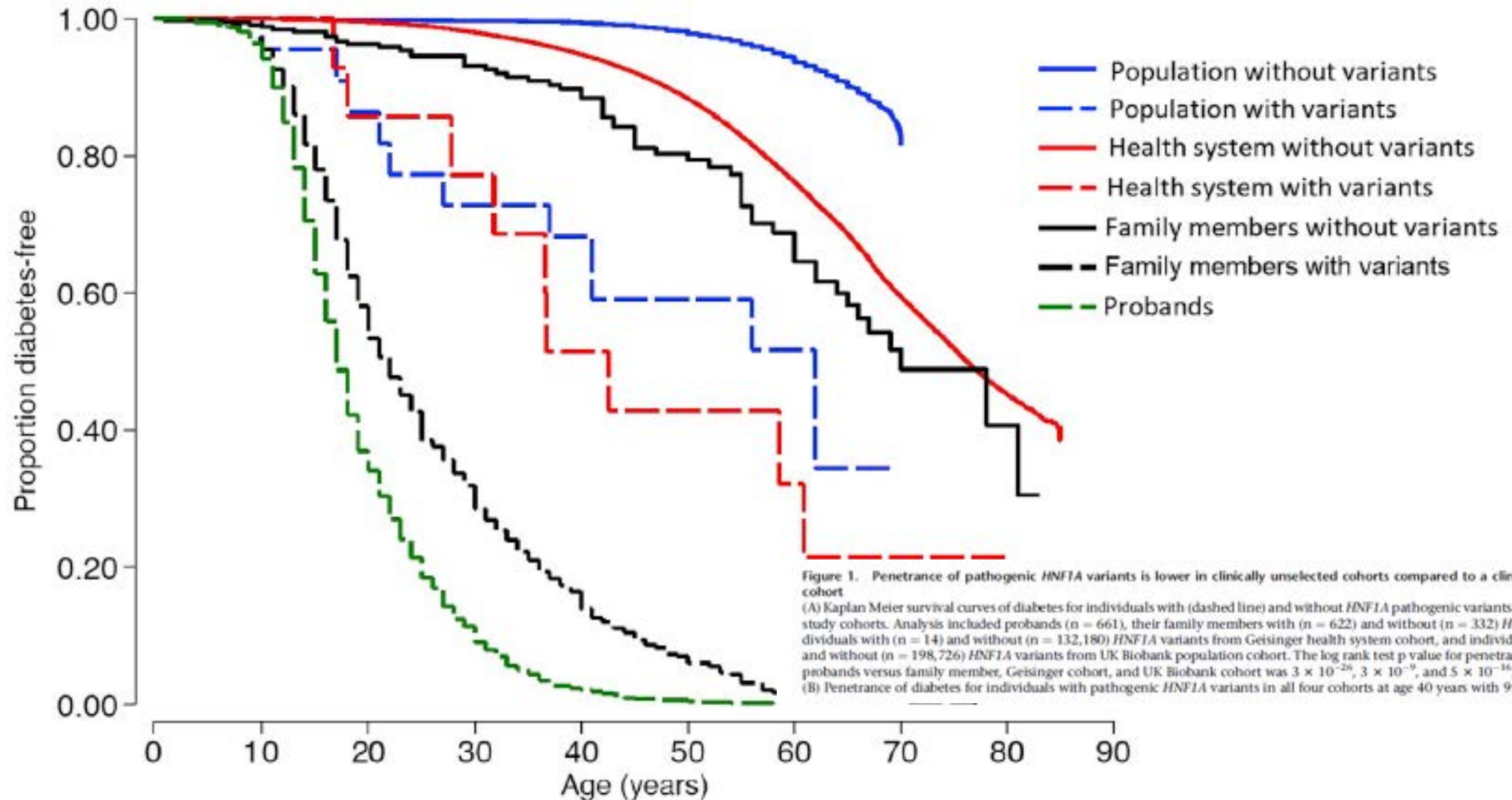
Estimation of genomic risk (penetrance): population-based case-control studies

	<i>BRCA1</i>	<i>BRCA2</i>	<i>PALB2</i>	<i>CHEK2</i>	<i>ATM</i>
Frequency in 'unselected' BC	0.99% 1 in 101	1.48% 1 in 68	0.54% 1 in 187	1.36% 1 in 73	0.76% 1 in 132
Association	OR=8.73 (7.47-10.20)	OR=5.68 (5.13-6.30)	OR=4.3 (3.68-5.03)	OR=2.4 (2.21-2.62)	OR=2.16 (1.93-2.41)
Subtype Association	TNBC ✓✓✓✓	TNBC ✓✓	TNBC ✓✓	ER+ ✓✓✓✓	ER+ ✓✓✓✓

- Meta-analysis of BRIDGES, CARRIERS, UKBiobank
- **100,093** female BC cases (**unselected**, population-type)
- **314,253** controls (unaffected females)

	<i>BARD1</i>	<i>RAD51C</i>	<i>RAD51D</i>	<i>TP53</i>	<i>PTEN</i>	<i>NF1</i>	<i>STK11</i>	<i>CDH1</i>
Frequency in 'unselected' BC	0.15% 1 in 672	0.11% 1 in 913	0.09% 1 in 1,079	0.05% 1 in 1,844	0.03% 1 in 3,755	0.07% 1 in 1,470	0.01% 1 in 11,525	0.04% 1 in 2,668
Association	OR=2.34 (1.85-2.97)	OR=1.53 (1.15-2.04)	OR=1.76 (1.15-2.41)	OR=3.62 (1.98-6.61)	OR=2.63 (1.38-5.02)	OR=2.01 (1.43-2.83)	OR=1.60* (0.48-5.30)	OR=2.01 (1.25-3.24)
Subtype Association	TNBC ✓✓✓	TNBC ✓✓✓	TNBC ✓✓✓					

Penetrance for HNF1A vars for diabetes



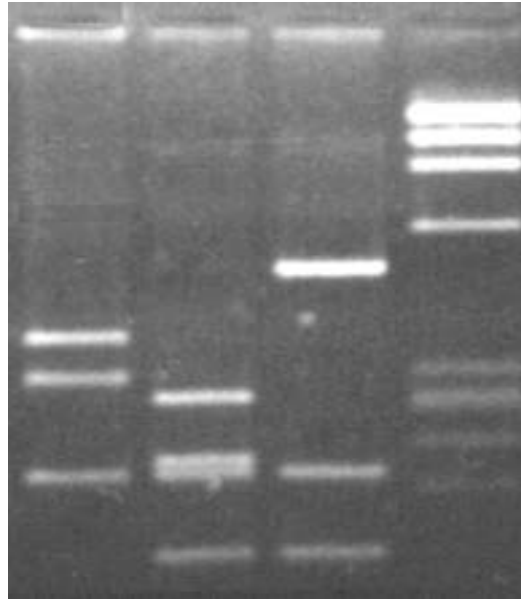
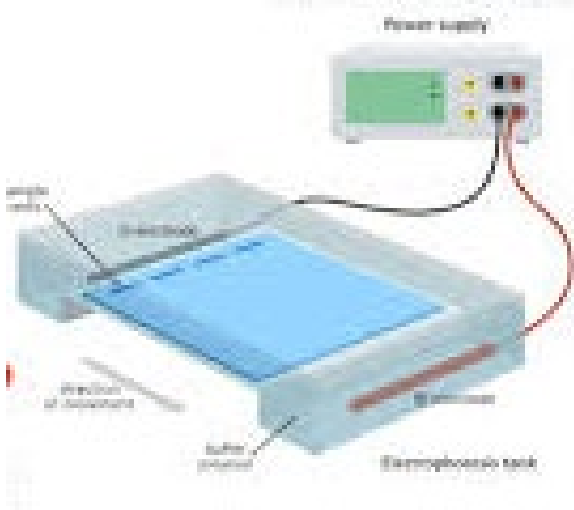


Phenotypes and ascertainment: The impact of technology

Dramatic Sequencing Technology shift

Analyzing for gene mutations: 2007

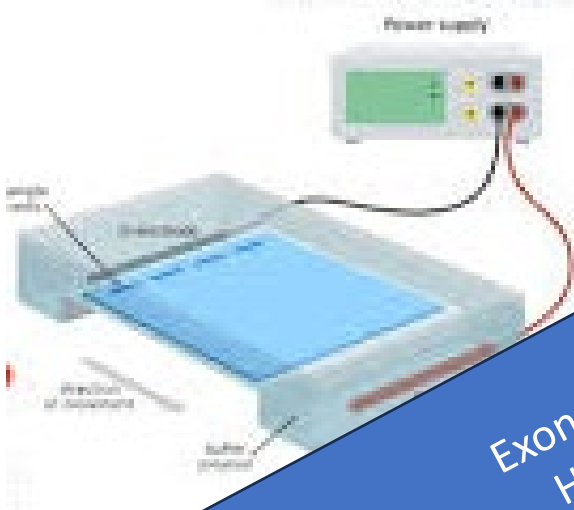
Conformation sensitive gel-electrophoresis



Dramatic Sequencing Technology shift

Analyzing for gene mutations: 2007

Conformation sensitive gel-electrophoresis



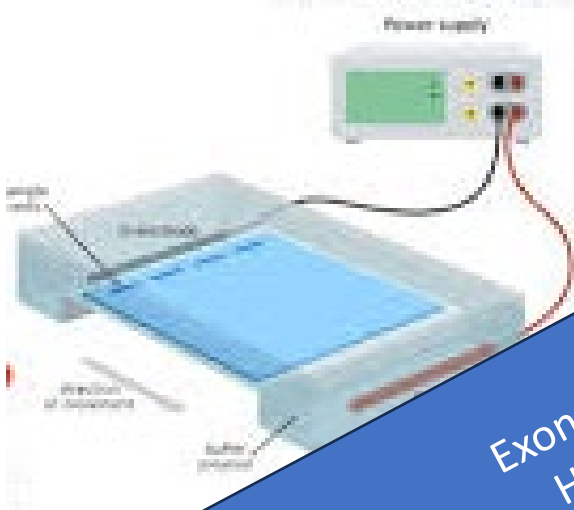
Exon-by-exon
High cost
Laborious
Think carefully about each gene tested



Dramatic Sequencing Technology shift

Analyzing for gene mutations: 2007

Conformation sensitive gel-electrophoresis

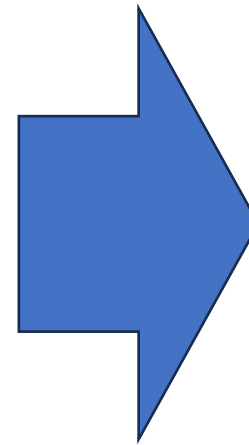
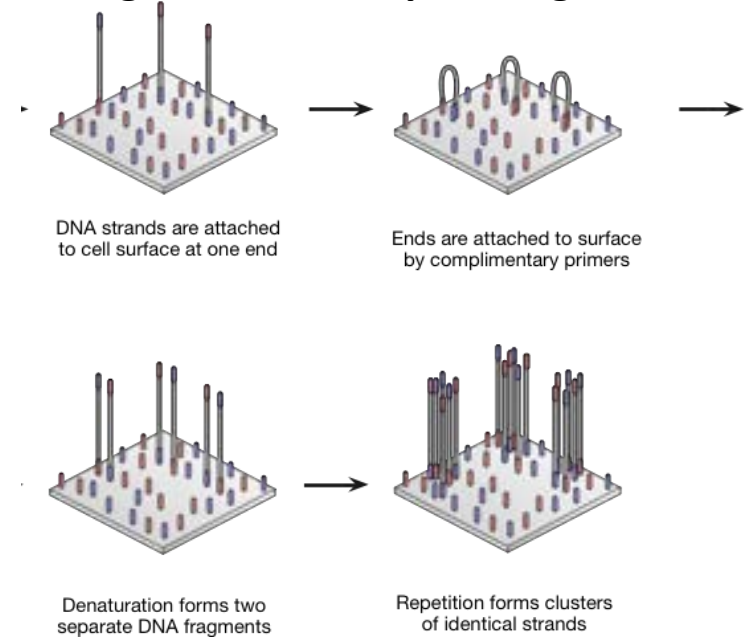


Exon-by-exon
High cost
Laborious
Think carefully about each gene tested



Analyzing for gene mutations: 2012

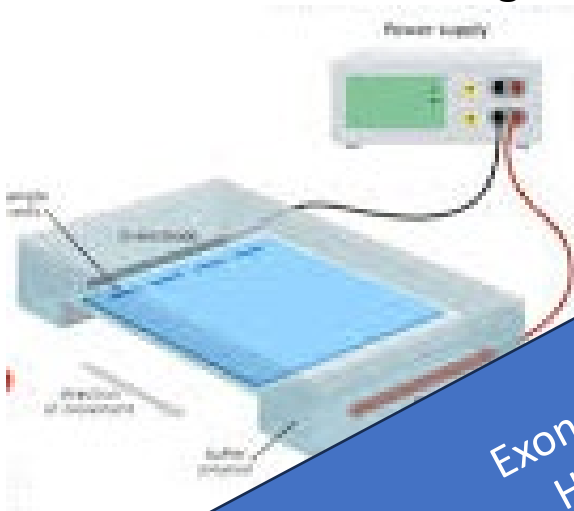
Next generation Sequencing



Dramatic Sequencing Technology shift

Analyzing for gene mutations: 2007

Conformation sensitive gel-electrophoresis

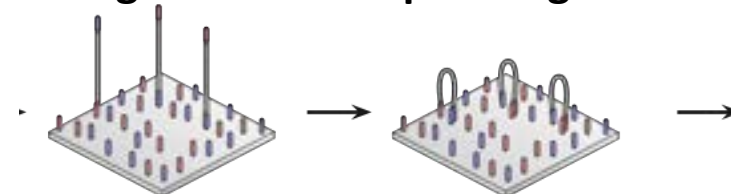


Exon-by-exon
High cost
Laborious
Think carefully about each gene tested



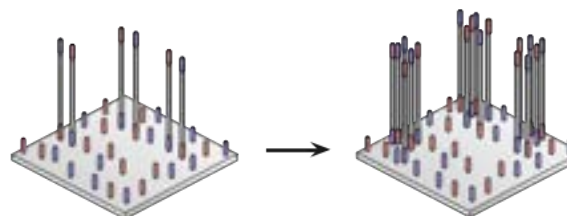
Analyzing for gene mutations: 2012

Next generation Sequencing



DNA strands are attached to cell surface at one end

Ends are attached to surface by complimentary primers



Denaturation forms two separate DNA fragments

Repetition forms clusters of identical strands

Analyzing for gene mutations: 2025

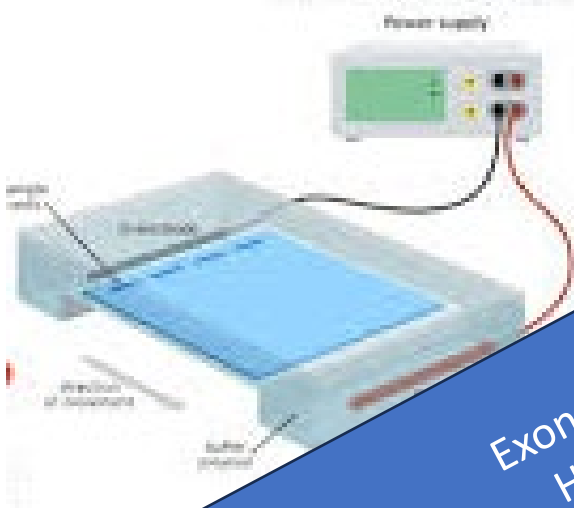
Next generation Sequencing



Dramatic Sequencing Technology shift

Analyzing for gene mutations: 2007

Conformation sensitive gel-electrophoresis

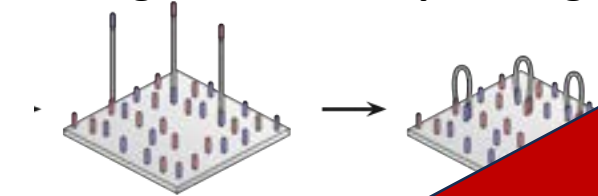


Exon-by-exon
High cost
Laborious
Think carefully about each gene tested



Analyzing for gene mutations: 2012

Next generation Sequencing



DNA strands are attached to cell surface at one end

wetlab cost equivalent for 2 vs 200 genes
running fewer very large panels more efficient

Repetition forms clusters of identical strands

Analyzing for gene mutations: 2025

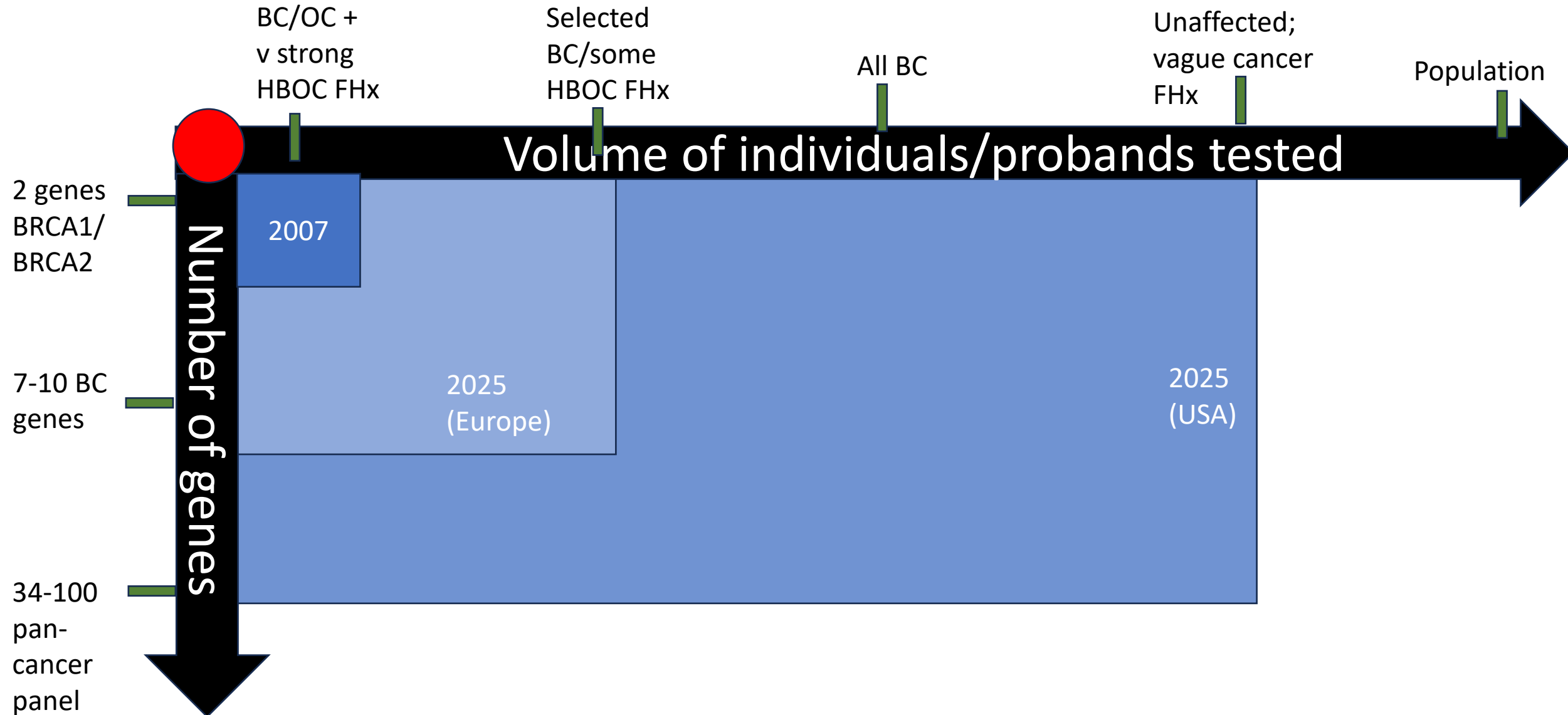
Next generation Sequencing





Phenotypes and ascertainment: The impact of technology Breast cancer as an example

Expansion of germline genetic/genomic testing: Breast(-Ovarian) cancer susceptibility as exemplar

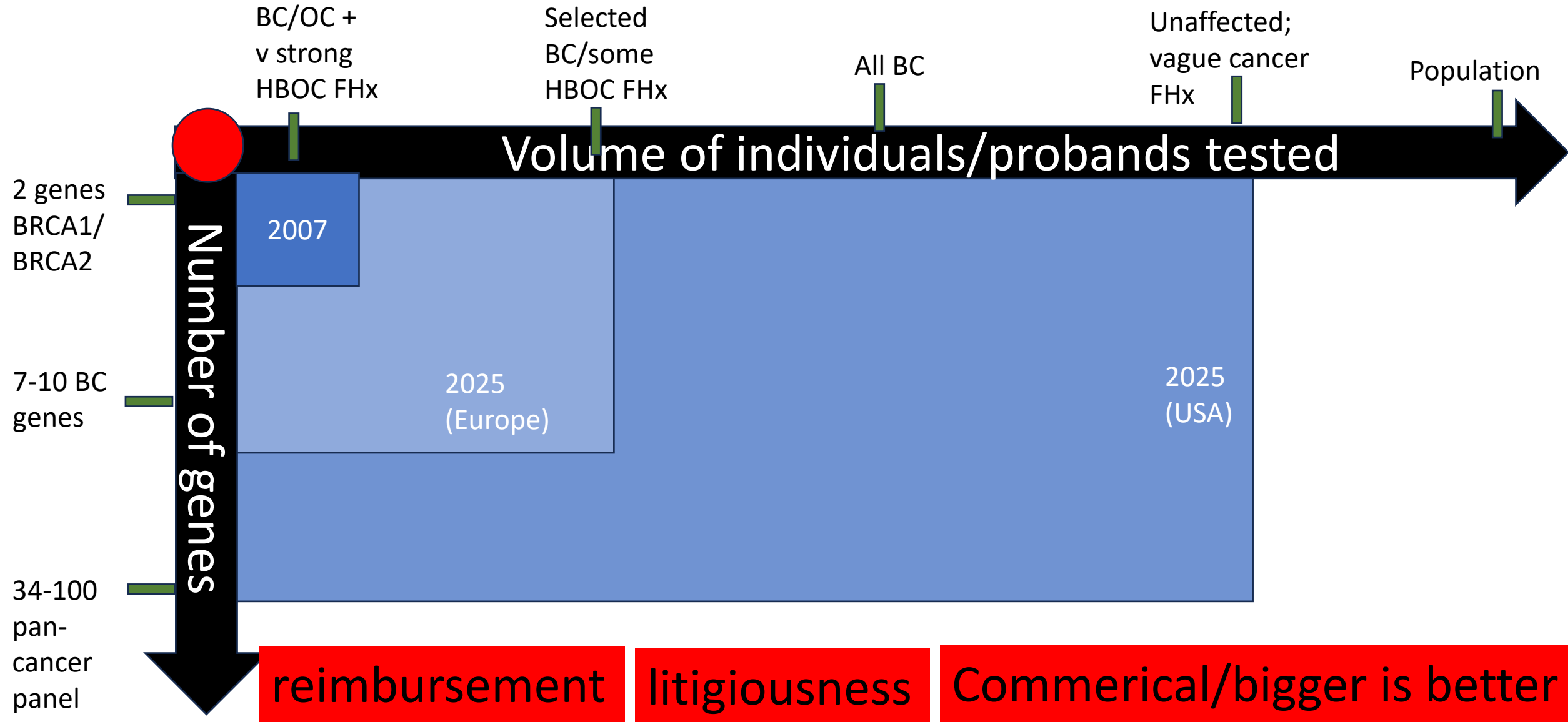


Expansion of germline genetic/genomic testing:

Breast(-Ovarian) cancer susceptibility as example



NCCN guidance



Complexities/challenges: Phenotype and ascertainment

Phenotypes are imperfectly captured

- **Subjectivity** of features
- Absence of objective **systems** for quantitation of phenotypic severity (some attempts a
- Poor (non-standardised) **documentation**
- **Age-related** penetrance
- Frequency/severity in general population poorly catalogued: **specificity**

Penetrance is uncertain

- Segregation studies based on **clinic-based phenotypically-driven** ascertainment: **very skewed and “traditional” estimates of phenotypic penetrance substantially over-estimated**
 - Patients only tested for the gene if they look to have the correct features: **self fulfilling prophecy**
 - Further **affirms** association between gene and features
- Population studies may exclude young onset/syndromic/lethal disease: **underestimate penetrance**

Testing context has shifted dramatically due to technology (esp WES/WGS/huge panels)

- Many individuals tested may have **minimal, unrelated or no phenotype**
- Phenotype may be poorly documented in test request (as test no longer needs to be justified)
- → “real-world clinical testing data” will be increasingly problematic as phenotype in which variant identified now highly variable

Themes

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3. Variant-specific risks/genotype-phenotype
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Genotype-Phenotype

Pathogenic Variants of **different types** or in **different gene regions** result in distinct clinical phenotypes

- ***FGFR3:***

- *different activating missense vars:*
 - Achondroplasia,
 - Thanatophoric dysplasia type II (lethal),
 - Hypochondroplasia

- ***LMNA — Lamin A/C:***

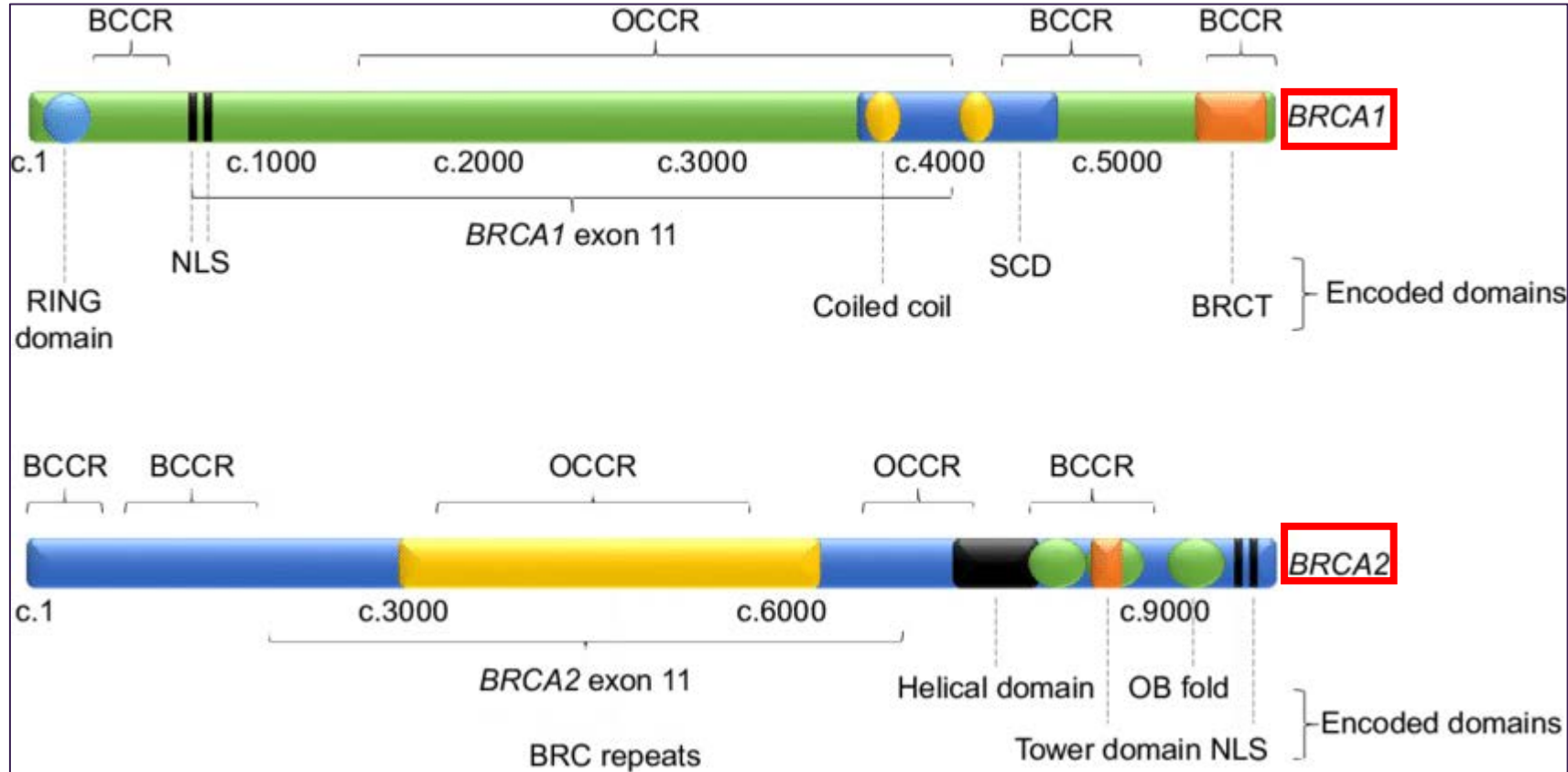
- *Missense vars in different domains:*
 - Emery–Dreifuss muscular dystrophy
 - Dilated cardiomyopathy with conduction disease
 - Hutchinson–Gilford progeria syndrome
- *Truncating variants*
 - severe congenital muscular dystrophy
 - metabolic laminopathies

- ***MECP2:***

- *PTV:* classic Rett syndrome;
- *Missense vars:* atypical Rett or isolated intellectual disability

Variant-specific risks

Protein-truncating mutations (ie conferring nonsense mediated decay)



Protein truncating mutations as a group, against case/control epi data

- OCCR: elevated risk of ovarian cancer, reduced risk of BC
- BCCR: elevated risk of BC, reduced risk of OC
- Reported 2001
- Reproduced 2015
- BRCA: strong epi data with large numbers, ?other genes
- ?splice-site variants (leakiness for rescue isoforms), ?true for other protein truncating variants

Variant-specific risks

Missense mutations

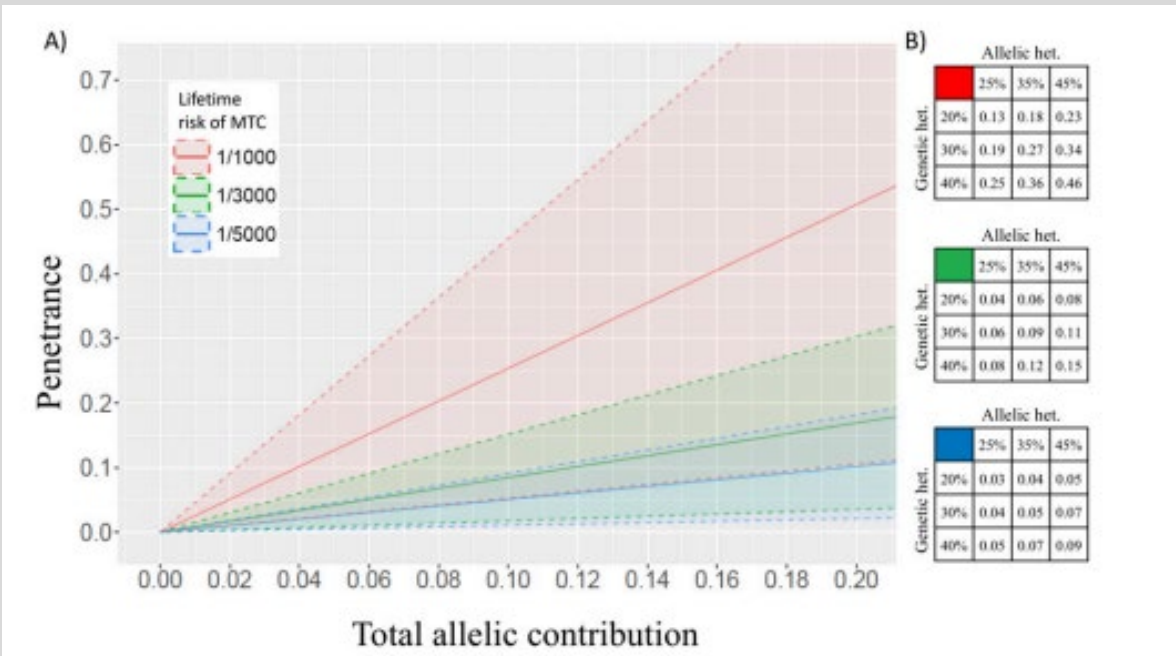
- Empirically: location and nature of missense mutation would confer different impact (=risk)
- BRCA1 c. 5096G>A p.R1699Q (↓): segregation analysis
- ATM c.7271T>G p.V2424G (↑): segregation analyses, case control
- RET c.2410G>A p.V804M (↓): observation in controls

Moghadasli, D, Meeks, H et al, *J Med Genet.* 2018
Southey, M, Goldgar, D et al *J Med Genet.* 2016

p.Val804Met, the Most Frequent Pathogenic Mutation in *RET*, Confers a Very Low Lifetime Risk of Medullary Thyroid Cancer

Chey Loveday,¹ Katherine Josephs,² Daniel Chubb,¹ Adam Gunning,³ Louise Izatt,⁴ Marc Tischkowitz,⁵ Sian Ellard,³ and Clare Turnbull^{1,4,6,7}

J Clin Endocrinol Metab, November 2018, 103(11):4275–4282



Loveday C, Josephs K, et al, *J Clin Endocrinol Metab* 2018
Whiffin N et al *Genetics in medicine* 2017

- RET mutations: **>90% penetrance** for medullary thyroid cancer (aggressive),
- Management:** prophylactic thyroidectomy in childhood
- For **p.Val804Met**, the lifetime penetrance for MTC, estimated from the allele frequency observed in non-TCGA ExAC population (11/55,798) was **4% [95% CI 0.9%-8.0%]**.

MTAF: maximum tolerated allele frequency

Penetrance for disease	Phenotype Frequency (general population)
Contribution (gene; novel variant in gene)	Variant frequency (general population)

Penetrance for HNF1A vars for diabetes

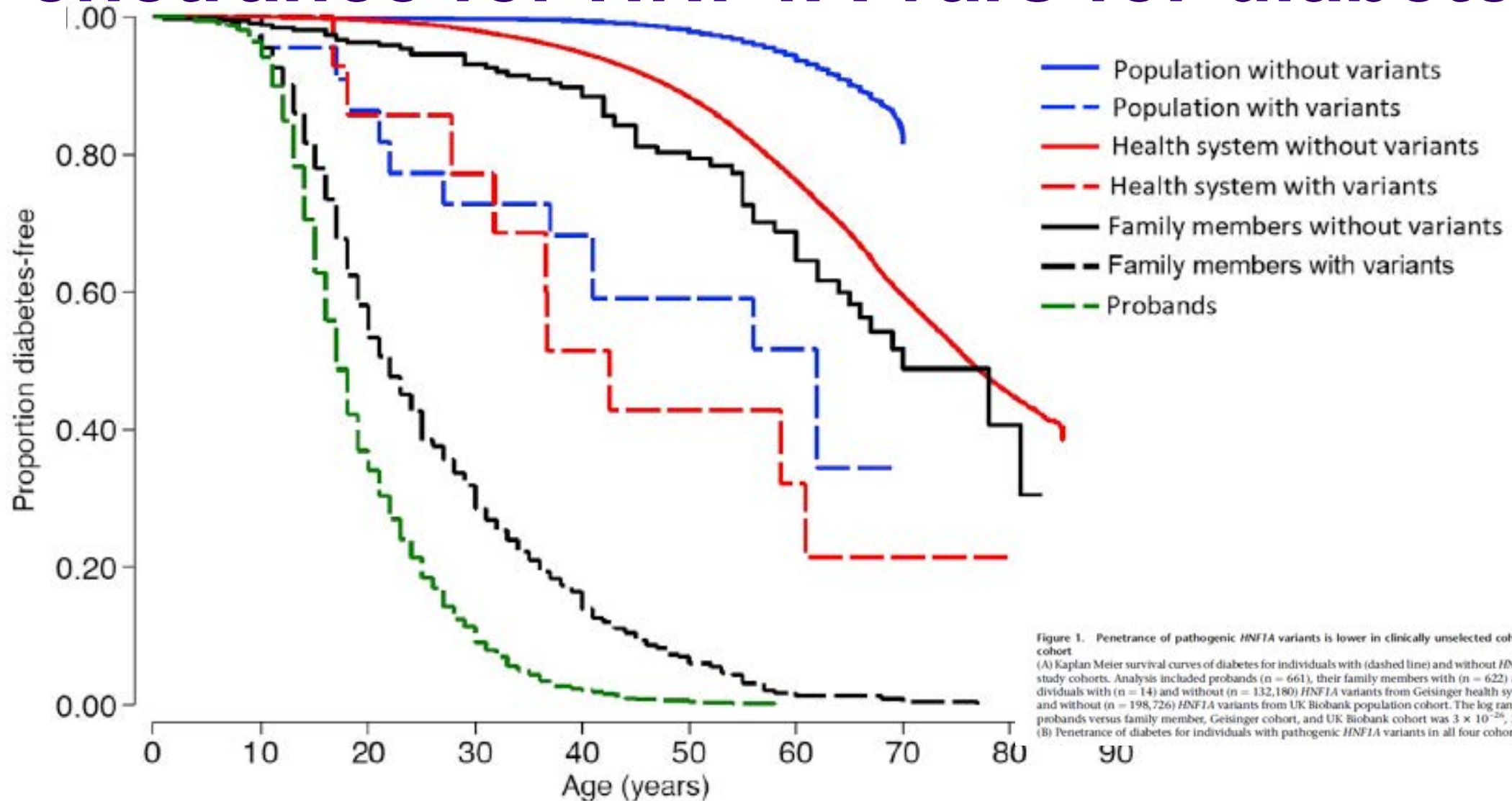
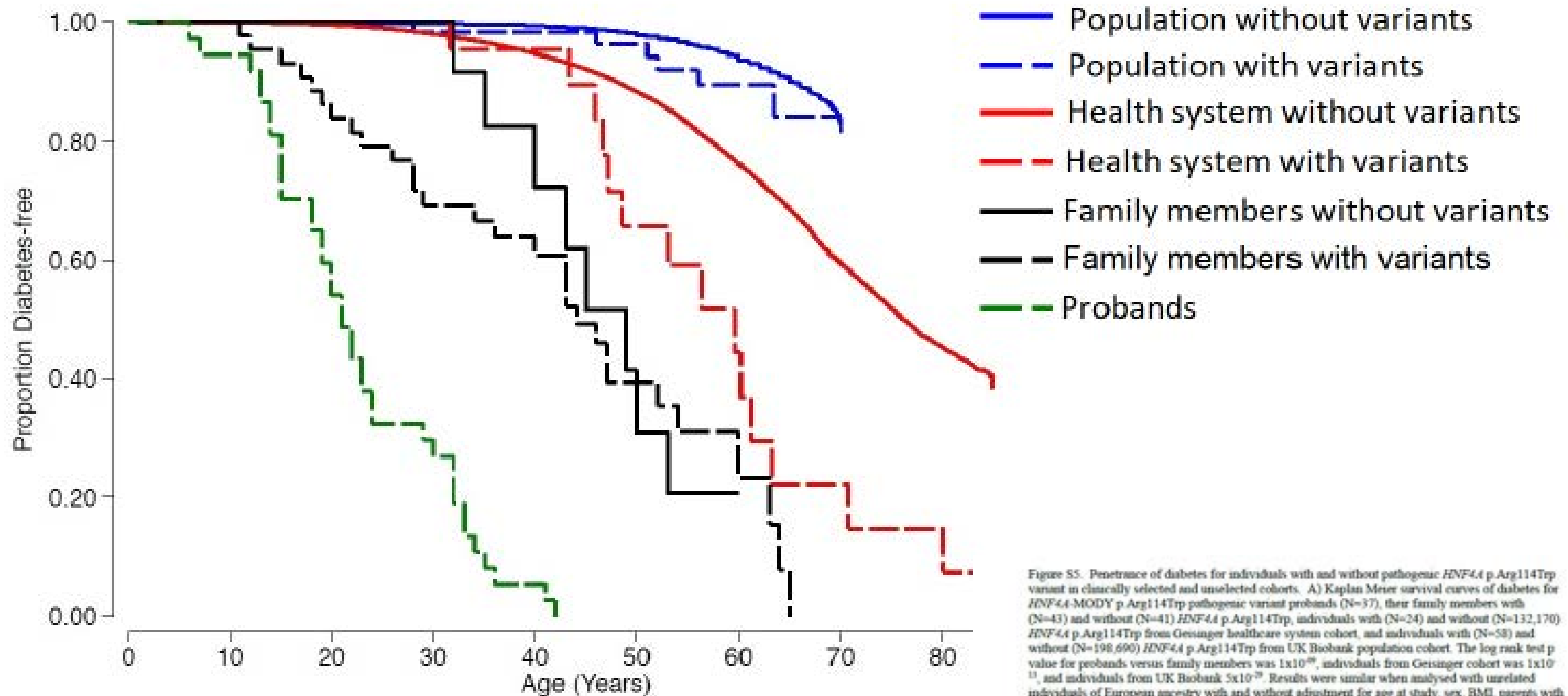


Figure 1. Penetrance of pathogenic *HNF1A* variants is lower in clinically unselected cohorts compared to a clinically ascertained cohort
 (A) Kaplan Meier survival curves of diabetes for individuals with (dashed line) and without *HNF1A* pathogenic variants (solid line) in four study cohorts. Analysis included probands ($n = 661$), their family members with ($n = 622$) and without ($n = 332$) *HNF1A* variants, individuals with ($n = 14$) and without ($n = 132,180$) *HNF1A* variants from Geisinger health system cohort, and individuals with ($n = 22$) and without ($n = 198,726$) *HNF1A* variants from UK Biobank population cohort. The log rank test p value for penetrance of diabetes for probands versus family member, Geisinger cohort, and UK Biobank cohort was 3×10^{-26} , 3×10^{-9} , and 5×10^{-16} , respectively.
 (B) Penetrance of diabetes for individuals with pathogenic *HNF1A* variants in all four cohorts at age 40 years with 95% CI.

Penetrance for HNF1A R114W for diabetes



Complexities/challenges: Variant-Specific Risks

- Variant-specific risks inevitable: esp missense and splicing
 - penetrance will also vary for each component phenotype, eg breast vs ovarian cancer
- Difficult to quantify intermediate penetrance
 - Variants individually rare: limited probands/families
 - Phenotypes binary (not continuous)
- Highly important for “clinical actionability”, as benefit/harm of clinical actions predicated on magnitude of disease risk.

→MAVES offer potential to address variant-specific risk

- Problem: need for multiple clinical positive controls to validate assay readout
 - to calibrate intermediate results (?accuracy of clinical risk, few examples)
- Require multiple replicates of MAVE
 - to demonstrate reproducibility of intermediate MAVE results

Themes

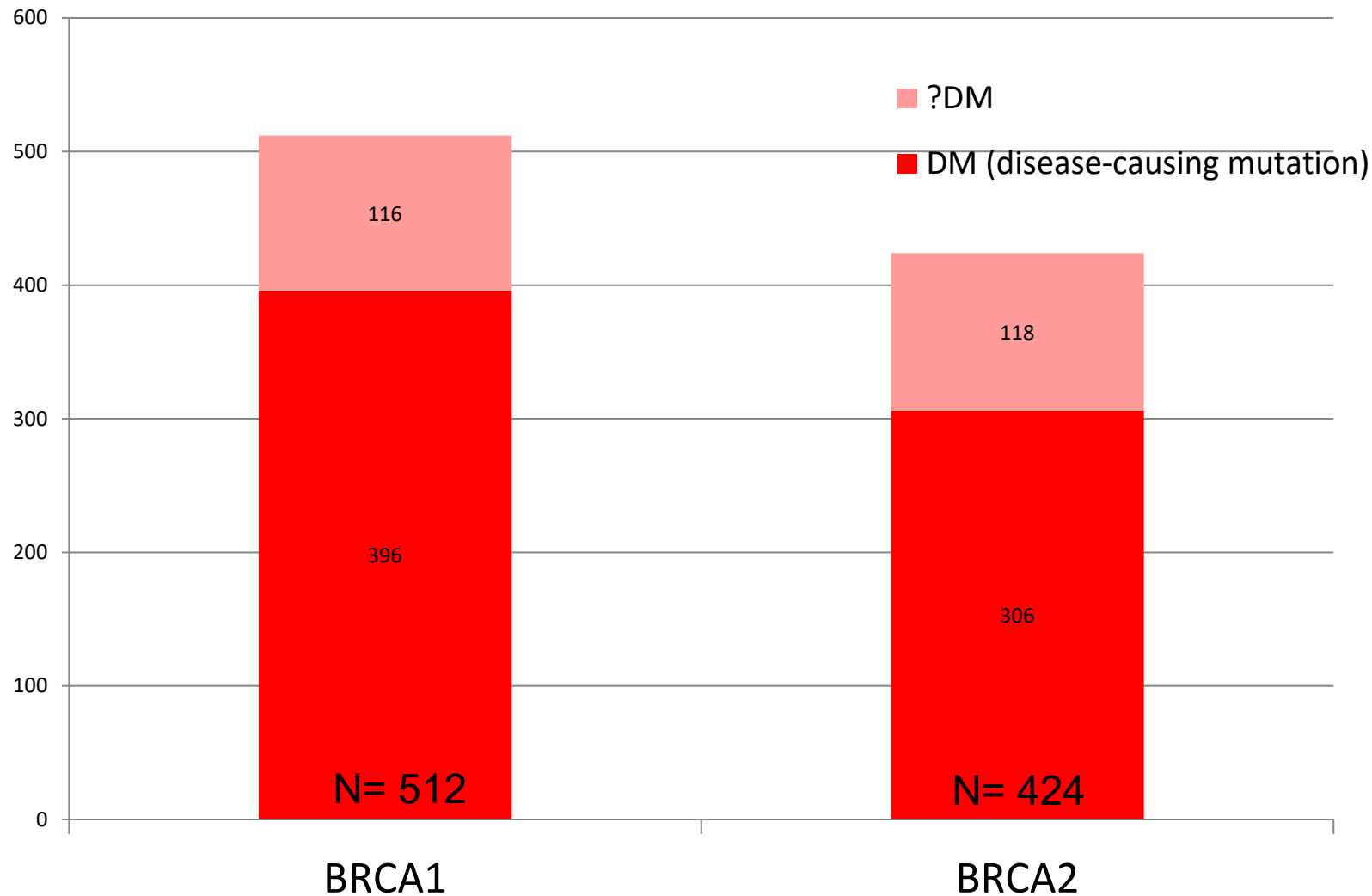
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Approaches/frameworks for variant classification

- 1) **Reductive**: Variant detected in disease→variant pathogenic
- 2) **Multifactorial**: statistical quantitation (centralised)
- 3) **ACMG v3.0 2015**: categorical framework (decentralised)
- 4) **ACMG v4.0 2026**: quantitative **AND** categorical framework (decentralised)

Classification based on presence of relevant phenotype

Human Gene Mutation Database: accessed November 2011



- 936 BRCA1/BRCA2 variants in HGMD in Nov 2011
- Hand-curated database of “clinically reported” variants (ie reported in medical literature)
- Any ‘rare’ variant reported in **case of disease** assigned as DM or ?DM
- Particularly flawed for common phenotypes (eg breast cancer)
- HGMD used as “clinical truthset” by computational biology community

Approaches/frameworks for variant classification

- 1) **Reductive**: Variant detected in disease→variant pathogenic
- 2) **Multifactorial**: statistical quantitation (centralised)
- 3) **ACMG v3.0 2015**: categorical framework (decentralised)
- 4) **ACMG v4.0 2026**: quantitative **AND** categorical framework (decentralised)

BRCA1/BRCA2 Multifactorial Analysis (2007)

A Systematic Genetic Assessment of 1,433 Sequence Variants of Unknown Clinical Significance in the *BRCA1* and *BRCA2* Breast Cancer–Predisposition Genes

Douglas F. Easton, Amie M. Deffenbaugh, Dmitry Pruss, Cynthia Frye, Richard J. Wenstrup, Kristina Allen-Brady, Sean V. Tavtigian, Alvaro N. A. Monteiro, Edwin S. Iversen, Fergus J. Couch, and David E. Goldgar

www.ajhg.org

The American Journal of Human Genetics Volume 81 November 2007

873

Table 3. BRCA Sequence VUSs with Odds of $\geq 20:1$ in Favor of Causality

Gene and Variant	Family-History LLR	LLR		Combined LLR	Odds in Favor of Causality
		Co-Occurrence	Cosegregation		
BRCA1:					
A1708E	11.22	1.10	.00	12.32	2,110,000,000,000
IVS20+1G→A	11.11	.52	.00	11.63	431,000,000,000
IVS13+1G→A	6.09	.60	.00	6.68	4,818,331
R1495M (4603G→T)	5.28	.81	.00	6.10	1,249,808
R1699W	4.34	.26	.00	4.60	39,978
G1788V	2.37	.42	1.06	3.85	7,054
IVS5+3A→G	3.04	.12	.00	3.15	1,417
G1706E	2.35	.13	.29	2.77	589

- Case-based data only
 - Co-occurrence
 - Co-segregation
 - Family history
- Subsequent addition of:
 - in-silico prediction
 - functional assays
- **Evolved multifactorial approaches:**
 - **ENIGMA (breast cancer)**
 - **INSIGHT (mismatch repair genes)**

Approaches/frameworks for variant classification

- 1) **Reductive**: Variant detected in disease→variant pathogenic
- 2) **Multifactorial**: statistical quantitation (centralised)
- 3) **ACMG v3.0 2015**: categorical framework (decentralised)
- 4) **ACMG v4.0 2026**: quantitative **AND** categorical framework
(?decentralised)
(decentralised)



	Strong	Supporting	Supporting	Moderate	Strong	Very strong
Population data	MAF is too high for disorder BA1/BS1 OR observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data →		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in trans with a dominant variant BP2 Observed in cis with a pathogenic variant BP2		For recessive disorders, detected in trans with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

ACMG v3.0 framework (2015)

Genetics in Medicine

ACMG STANDARDS AND GUIDELINES

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,3}, Sherri Bale, PhD⁴, David Bick, MD⁵, Soma Das, PhD⁶, Julie Gastier-Foster, PhD^{1,2,3}, Wayne W. Grody, MD, PhD^{1,2,3,7}, Madhuri Hegde, PhD^{1,3}, Elaine Lyon, PhD^{1,3}, Elaine Spector, PhD^{1,3}, Karl Voelkerding, MD^{1,3} and Heidi L. Behm, PhD^{1,3}; on behalf of the ACMG Laboratory Quality Assurance Committee

Table 5 Rules for combining criteria to classify sequence variants

Pathogenic	(i) 1 Very strong (PVS1) AND (a) ≥1 Strong (PS1–PS4) OR (b) ≥2 Moderate (PM1–PM6) OR (c) 1 Moderate (PM1–PM6) and 1 supporting (PP1–PP5) OR (d) ≥2 Supporting (PP1–PP5) (ii) ≥2 Strong (PS1–PS4) OR (iii) 1 Strong (PS1–PS4) AND (a) ≥3 Moderate (PM1–PM6) OR (b) 2 Moderate (PM1–PM6) AND ≥2 Supporting (PP1–PP5) OR (c) 1 Moderate (PM1–PM6) AND ≥4 supporting (PP1–PP5)
Likely pathogenic	(i) 1 Very strong (PVS1) AND 1 moderate (PM1–PM6) OR (ii) 1 Strong (PS1–PS4) AND 1–2 moderate (PM1–PM6) OR (iii) 1 Strong (PS1–PS4) AND ≥2 supporting (PP1–PP5) OR (iv) ≥3 Moderate (PM1–PM6) OR (v) 2 Moderate (PM1–PM6) AND ≥2 supporting (PP1–PP5) OR (vi) 1 Moderate (PM1–PM6) AND ≥4 supporting (PP1–PP5)
Benign	(i) 1 Stand-alone (BA1) OR (ii) ≥2 Strong (BS1–BS4)
Likely benign	(i) 1 Strong (BS1–BS4) and 1 supporting (BP1–BP7) OR

Issues with ACMG v3.0 (2015) framework

1) Loose definitions/terminology

- Requirement for additional specification: ClinVar SVI/VCEPs, CanVIG-UK

2) Combinations of evidence items:

- Multiple data items non-orthogonal

3) Categorical, unscaled, empirical.

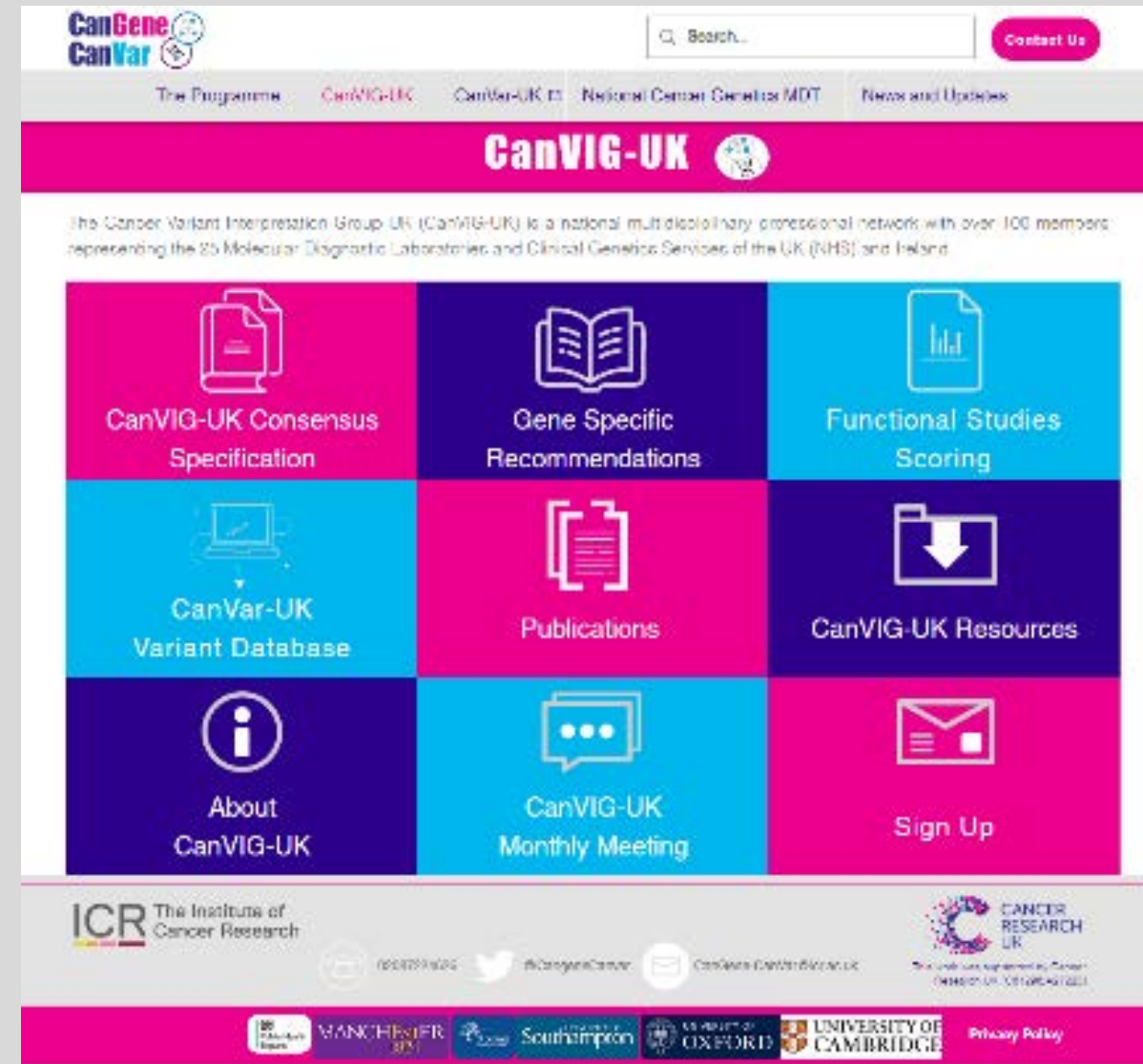
4) Universal vs disease-specific evidence items

- Variant Curation Expert Panels ('VCEPs') can vary in
 - How they define/interpret **universal** evidence elements
 - strictness for disease-specific evidence

Cancer Variant Interpretation Group (CanVIG-UK)

- Monthly meetings since 2017
- 330 Members: NHS Clinical Scientists, Consultants, Clinical Geneticists, Genetic Counsellors
- Specification of ACMG guidance (consensus)
- (Interim) gene-specific guidance
 - Specification of VCEP guidance
- Guidance on broad issues eg:
 - Reduced penetrance classification
- Platform for information sharing: CanVar-UK
- Education and training

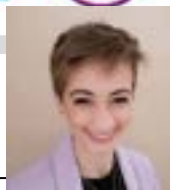
<https://www.cangene-canvaruk.org/about-canvig-uk>



<https://www.cangene-canvaruk.org/canvig-uk>



Alice Garrett



Sophie Allen

37th version



CanVIG-UK Consensus Specification for Cancer Susceptibility Genes (CSGs) of ACGS Best Practice Guidelines for Variant Classification

Date: 13/12/2024 Version: 3.00

A. Garrett^{1,2}, S. Allen¹, L. Loong¹, M. Durkie³, J. Drummond⁴, G.J. Burghell⁵, R. Robinson⁶, A. Callaway⁷, J. Field⁸, B. Frugthiet⁹, S. Palmer-Smith¹⁰, J. Grant¹¹, J. Pagan¹¹, T. McDavitt¹², T. McVeigh¹³, H. Hanson^{1,14,15}, C. Turnbull^{1,13}

2.18	05/05/2023	PS4	Clarification regarding use of PS4 where PM2 cannot be applied, UK Biobank replaced gnomAD v2.1.1 as recommended population control data source. Rewording of explanatory notes re: case-counting.	Allen/ Garrett	CStAG
2.18	05/05/2023	PVS1	Amended link to Tayoun et al. decision tree	Allen	CStAG
2.18	26/05/2023	PM2	Updated guidance on population database use and frequency thresholds.	Garrett/ Allen	CStAG
2.18	03/07/2023	PM5	Clarification of application of PM5 supporting when REVEL score <0.773.	Garrett	CStAG
2.18	04/07/2023	Combinations	PVS1- PM5 combination amended to "caution" due to gene-specific recommendations for protein truncating variants.	Garrett/ Allen	CStAG
2.18	15/09/2023	BA1/BS1	Specification regarding use of the gnomAD filtering allele frequency	Callaway/ Garrett	CSxAG
2.18	28/09/2023	PVS1	Clarification of PVS1 use for stop codon variants		

Widely used
internationally
and for GenQA

Theme: COMPUTATIONAL AND PREDICTIVE DATA

PVS1 (null variant): Null variant (nonsense, frameshift, canonical ± 1 or 2 splice sites, initiation codon, single or multi-exon deletion) in a gene where LOF is a known mechanism of disease

VSTR _MOD
_STR _SUP

For guidance on application of this criteria see Tayoun et al, 2018 (in particular the [PVS1 decision tree](#) and associated notes⁴) and the [ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2020 v4](#) (notes on PVS1 & figure 2)⁵.

PM2 (rare in controls): Absent from controls (or at extremely low frequency if recessive) in Exome Sequencing Project, 1000 Genomes Project, or ExAC

_MOD
_SUP

Use at **Moderate**: where 0 observations of the variant across all populations in the non-cancer portion of gnomAD and the entirety of UK Biobank

Theme: POPULATION DATA

BA1/BS1 (common in controls): Allele frequency is "too high" for disorder (Exome Sequencing Project, 1000 Genomes Project, or Exome Aggregation Consortium)

_SA _STR

For use in dominant conditions for alleles of standard penetrance:

Use **BA1** as **Stand_Alone** when the allele frequency in any ethnicity-specific subpopulation of >1000 individuals, or mixed population of >5000 individuals is:

- >1% for well characterised cancer susceptibility genes¹⁶ OR
- The Popmax filtering allele frequency (Popmax FAF) given in gnomAD is greater than the BA1 maximum tolerated allele frequency (MTAF) threshold specified for the specific gene by respective expert group (VCEP/CanVIG guidance documents). See explanatory notes below.

CanVIG Interim Gene-Specific Guidance

BRCA1/BRCA2: CanVIG-UK Gene-Specific Guidance

Date: 19/10/2023 Version: 1.19

MMR: CanVIG-UK Gene-Specific Guidance

Date: 23/10/2023 Version: 1.6

SDHB/D: CanVIG-UK Gene-Specific Guidance

Date: 24/03/22 Version: 1.3

RAD51C/RAD51D: CanVIG-UK Gene-Specific Guidance

Date: 28/09/2023 Version: 1.0

PM5: Novel missense change at an amino acid residue where a different missense change determined to be pathogenic seen before	_SUP	Per ATM VCEP guidance, PM5 may be applied at supporting level for truncating variants if nonsense mediated decay (NMD) is predicted. For <i>RAD51C</i> and <i>RAD51D</i> specific boundaries for NMD, please see below: <i>RAD51C</i>: NMD predicted if the variant is 5' of p.Leu326 <i>RAD51D</i>: gene-specific NMD boundary not yet established; assume NMD unless in the final exon or the last 50 base pairs of the penultimate exon.
BA1/BS1: Allele frequency is "too high" in ExAC or gnomAD for disorder	_SA _STR	<i>RAD51C</i> draft VCEP guidance states MTAf of: BA1: 0.000583 (0.0583%); BS1: 0.0000583 (0.00583%) Use these MTAfs for <i>RAD51C</i> and <i>RAD51D</i> .

And review of ClinGen VCEP guidelines:

CanVIG-UK review of APC November 2023: Consensus to use relevant recommendations from the **ClinGen InSiGHT Hereditary Colorectal Cancer/Polyps VCEP guidance for APC v2.0.3** (available at: <https://clinicalgenome.org/affiliation/50099/>, PDF attached below). Additional points of specification are given below where applicable.

Summary: Evidence towards Pathogenicity

Evidence element	Evidence strengths allowed				Thresholds/data-sources/applications specifically relevant to APC
PS4	_VSTR	_STR	_MOD	_SUP	As per VCEP
PP4					Not applicable as per VCEP
PM2				_SUP	As per VCEP
PVS1	_VSTR	_STR	_MOD	_SUP	As per VCEP
PS1		_STR	_MOD		As per VCEP
PM4					Not applicable as per VCEP
PM5			_MOD	_SUP	As per VCEP
PP3				_SUP	As per VCEP
PM1 and PP2					Not applicable as per VCEP
PS3	_VSTR	_STR	_MOD	_SUP	As per VCEP
PP1		_STR	_MOD	_SUP	As per VCEP
PS2 and PM6	_VSTR	_STR	_MOD	_SUP	As per VCEP
PM3					Not applicable as per VCEP
PP5					Not applicable as per VCEP

Summary: Evidence towards Benignity

BA1/BS1	_SA	_STR		As per VCEP
BS2		_STR	_SUP	As per VCEP
BP4			_SUP	As per VCEP
BP1			_SUP	As per VCEP
BP7			_SUP	As per VCEP
BP3				Not applicable as per VCEP
BS3		_STR	_SUP	As per VCEP
BS4		_STR	_SUP	As per VCEP
BP2			_SUP	As per VCEP
BP6				Not applicable as per VCEP
BP5			_SUP	As per VCEP

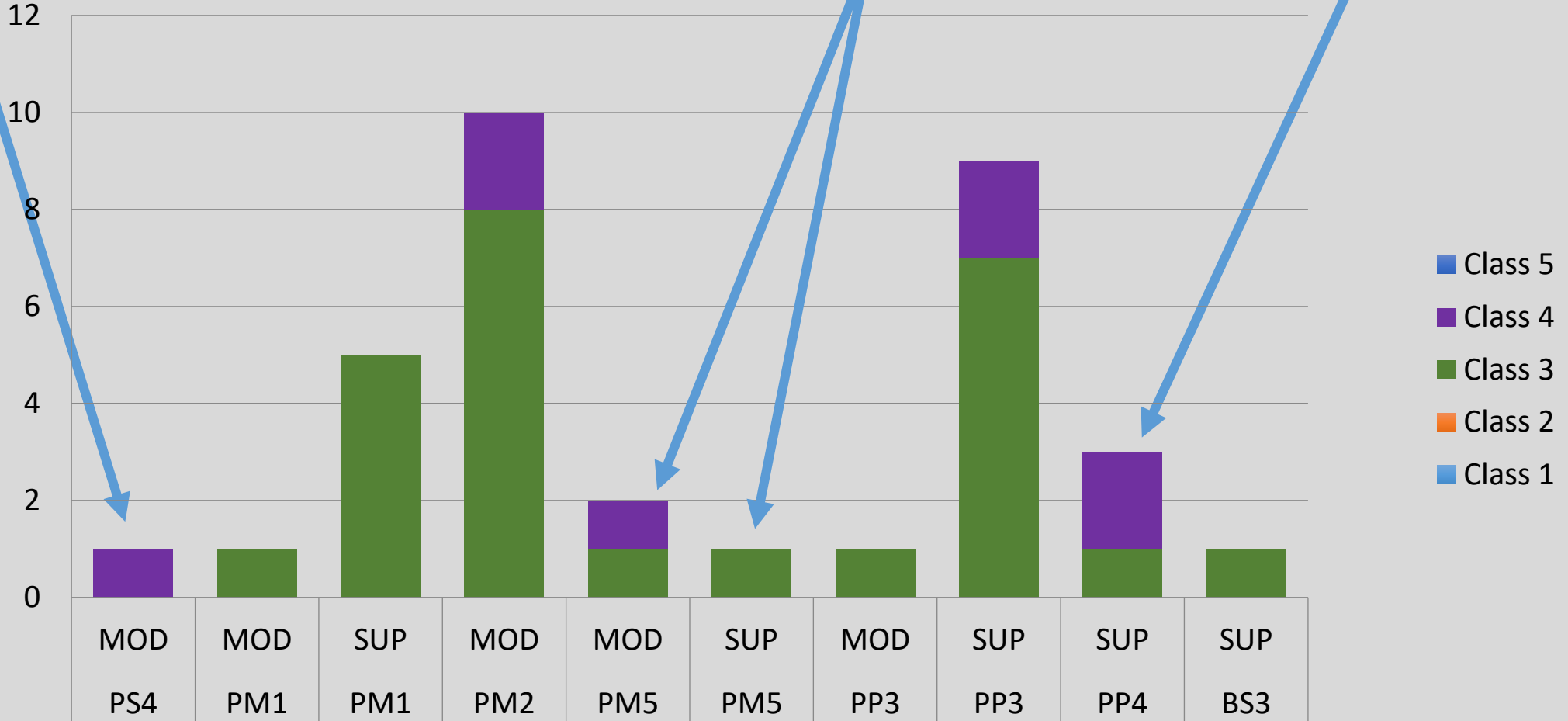
Reported on ClinVar x6 = VUS, LOVD x5 = VUS,
 UMD/BRCA Share x8 = VUS; Reported by Caputo *et al*
 2012; Reported by Coulet *et al* 2010 = **>20 cases**
 Can't be sure no overlap however still fairly compelling
 case-control evidence with **0 cases in gnomAD**
 Use of PS4 in absence of case-control data?

CanVIG-UK:

Variant(s) of the month

Can you use PM5 if you
 have functional data?

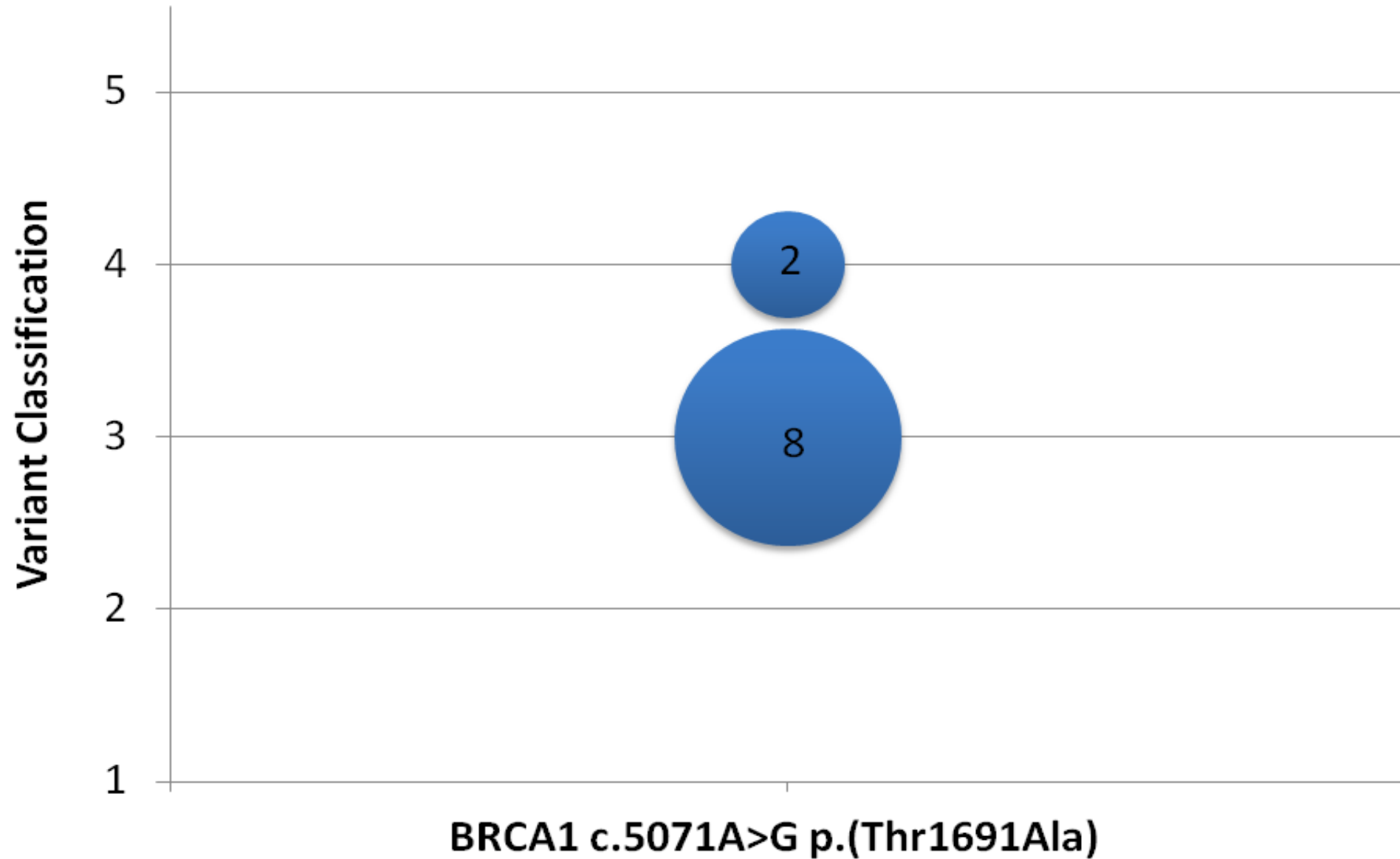
Use of PP4 in
 HBOC?



BRCA1 c.5071A>G p.(Thr1691Ala)

CanVIG-UK:

Variant(s) of the month



CanVar-UK: platform supporting variant classification

[LOGIN](#) [REGISTER](#)



Select a gene:

— Select Gene —

[GENE SUMMARY](#)

[MODEL COUNTS \(UKB\)](#)

Type a variant:

Type a variant

Try out CanVar-UK with these example variants:

- BRCA1 c.5297T>C p.(Val1736Ala)
- MSH2 c.1915C>T p.(His639Tyr)

Welcome to CanVar-UK, the interactive database for variant interpretation in cancer susceptibility genes

New to CanVar-UK? Start here!

Check out our [introductory video](#) which explains:

- What CanVar-UK is
- How to search for a variant
- What resources are available
- How to register for membership
- How to use the diagnostic user forum

New CanVar features

- Addition of **23 new cancer susceptibility genes**, with particular expansion of genes implicated in **hereditary** syndromes
- **UK Biobank exome allele frequencies** across **multiple ethnicities** for all CanVar SNVs and **small insertions-deletions**
- Addition of **tip bobble** for *in silico* and functional datasets, to facilitate their inclusion in variant interpretation
- Addition of new **functional** and **epidemiological** datasets from **Caputo et al. (2021)** and **Wai et al. (2020)**
- A **diagnostic user forum** for sharing **classifications**, **data**, and forum **email communication**

General CanVar features

- Data from functional, splicing and genetic epidemiology studies
- Results from various protein and splicing impact *in silico* tools
- Live links to ClinVar, and links to other external resources, such as VarSome, BRCA Exchange and cBioPortal
- Data from control series such as gnomAD
- Observation frequency in cases from UK diagnostic laboratories
- Gene summary-level data, including CanVar-UK classification guidance and exon start/end positions
- A custom tool for applying the PMS criterion of the ACMG guidance

Our resources are mostly available without registration; however, registering for CanVar-UK membership grants additional access to our diagnostic user forum, where members can post questions and/or email out to the CanVar-UK diagnostic user community, as well as view variant-level discussions and provisional classifications.

Registration is available to anyone whose work involves or relates to clinical diagnostics. During your registration, please include details about your institution and your role, and sign up using your work email address.

If you are an NHS scientist or clinician, please register with your NHS email address and your registration will be automatically approved. For other users, there may be a short wait while the CanVar-UK team reviews your application.

For general enquiries, feedback, or suggestions for new gene-specific or pan-gene resources/studies, please contact CanVarUK@icr.ac.uk.

For variant-specific information, papers or resources, please create a post on the diagnostic user forum page for that variant.

Disclaimer

The Cancer Variant Database is a repository of annotations for variants in cancer predisposition genes only.

The information on CanVar-UK is not intended for direct diagnostic use or medical decision-making without review by a genetics professional. Information displayed on CanVar-UK has not been independently verified.

<https://canvaruk.org/>

CanVar-UK: UK lab variant counts/control counts

BRCA1 c.5207T>C p.(Val1736Ala)

[Download a variant report](#) [Click to search another variant](#)

Summary

In Silico Predictions

Control/Case Frequency

Genetic Epidemiology

Functional/Splicing Analysis

CanVar-UK Diagnostic
Discussion Forum

Control Frequency

1 gnomAD Population Frequencies (v2.1.1 - Jan. 2019) [View](#)

- ☐ gnomAD v2.1.1
☒ gnomAD v2.1.1 non-cancer
☐ gnomAD v2.1.1 non-cancer female

	Allele Count	Homozyg.	Alleles sequenced	Individuals sequenced	WES in full dataset
European (Non-Finnish)	0	0	-	-	51377
Latino	0	0	-	-	17130
South Asian	0	0	-	-	15263
European (Finnish)	0	0	-	-	10816
East Asian	0	0	-	-	8846
African	0	0	-	-	7451
Ashkenazi Jewish	0	0	-	-	4706
Other	0	0	-	-	2510
Total	0	0	-	-	118479

1 UK Biobank Population Frequencies (retrieved Jan. 2023)

[Frequencies of non-base substitution small variants in UK Biobank](#)

- ☒ UK Biobank (all participants)
☐ UK Biobank (female-only)

	Allele Count	Homozyg.	Alleles sequenced	Individuals sequenced	WES in full dataset
White	10	0	884510	442259	442266
Asian	0	0	18182	9081	9081
Black	0	0	14534	7267	7267
Chinese	0	0	2904	1452	1452
Mixed	0	0	5390	2695	2695
Other	0	0	11538	5769	5769
Total	10	0	937048	468523	468530

Case Frequency

1 UK diagnostic labs (Release date: July 2023)

[Frequencies of non-base substitution small variants in NCRS GLH data](#)

Birmingham	3
Bristol	0
Cambridge	1
GOSH	0
Guy's	1
KGC	0
Leeds	0
Liverpool	0
Newcastle	3
Manchester	3
Newcastle & Leeds	0
Nottingham	1
Oxford	0
Royal Marsden	0
Salisbury	2
Sheffield	0
St. George's	2
Total proband count	16
Total probands tested	88722
Self-reported ethnicity: White (use as NFE): total proband count	16
Self-reported ethnicity: White (use as NFE): total probands tested	62319

1 Database Sources

	Number of times reported
LOVD3 (2019)	18
BIC (2015)	18
OMuDB	11

<https://canvaruk.org/>

CanVar-UK: in silico predictions/resources

BRCA1 c.5207T>C p.(Val1736Ala)

[Download a variant report](#) [Click to search another variant](#)

Summary

In Silico Predictions

Control / Case Frequency

Genetic Epidemiology

Functional / Splicing Analysis

CanVar UK Diagnostic Discussion Forum

Summary

Genome version	hg19
Gene	BRCA1
Ensembl transcript	ENST00000357654
RefSeq transcript	NM_007294.3
Chromosome	17
Genomic position (hg19)	41209138
Genomic position (hg38)	43657122
Reference allele	T
Variant allele	C
HGVS cDNA	c.5207T>C
HGVS (protein)	p.Val1736Ala
HGVS (protein, 1-letter)	p.V1736A
Exon	exon 20
rsID	rs45553935
Exon/Intron	exon 19 of 23
Variant location	exon
Variant type	substitution
Variant coding effect	missense
cDNA position	5207
Intronic position	N/A
Codon number	1736
First/Last 3 bases of exon	-

Literature Links

Google Scholar

Google Search

PubMed

Database Links

ClinVar

VarSome

LOVD

cBioPortal

InSiGHT

Variant	Interpretation	Review Status	Last Evaluated	
c.5207T>C	Pathogenic	★★★★☆	2019/06/25 00:00	
>> PM5: other variants at this codon (link retrieval may take a few seconds)				
Variant	Protein Change	ClinVar Interpretation	ClinVar Status	REVEL score
c.5206C>T	p.V1736V	not provided	☆☆☆☆	
c.5206C>G	p.V1736V	not provided	☆☆☆☆	
c.5206C>A	p.V1736V	not provided	☆☆☆☆	
c.5207T>G	p.V1736Q	Pathogenic/Likely pathogenic	★★★★☆	0.058
c.5207T>A	p.V1736D	not provided	☆☆☆☆	0.837
c.5206G>C	p.V1736L	not provided	☆☆☆☆	0.504
c.5206G>A	p.V1736I	not provided	☆☆☆☆	0.608
c.5206G>T	p.V1736F	Likely pathogenic	★★☆☆	0.794

Other Classifications

HMMD classification (2015)	DM
DMuGB classification (2015)	Unclassified/Not Known/Probably Pathogenic
UMD classification (2015)	Not found

https://canvaruk.org/

CanVar-UK: Functional Assay Data

[Summary](#)[In Silico Predictions](#)[Control / Case Frequency](#)[Genetic Epidemiology](#)[Functional / Splicing Analysis](#)[CanVar-UK Diagnostic Discussion Forum](#)

Functional Analysis

[Click here to view the current evidence statements applicable for functional assays, as quantified by CanVG-UK](#)

[Bouman et al. Cancer Discov. 2013](#)

Mutation type	VUS
AlignGVGD	C65
Predicted splice effect	No
Literature predictions	No consensus
Growth	+/-
IC50	-0.42 to 0.25
Classification	Not clear

1 [Saturation genome editing BRCA1 haploid cell assay \(Findlay 2018\)](#)

Function score	-1.6030525993
Function probability	0.9999167107
Functional classifications	LOF
Mean RNA score	-0.1588668542

1 [Fernandes et al., J Biol Chem. 2019](#)

Functional class (IClass)	5
Functional class (IClass) category	Pathogenic

1 [Bouman et al. Cancer Res. 2020](#)

Probability deleterious (cisplatin)	0.769017624
Prediction (cisplatin)	Not clear
Probability deleterious (olaparib)	0.9999999999999998
Prediction (olaparib)	Deleterious
Probability deleterious (DR-GFP)	1

Splicing Analysis

UK diagnostic labs (Wessex data) 1

Specimen	W1214516
Splice abnormality	None
Sample type	PAXGene

[Wai et al. Genetics in Medicine 2020](#)

Variant position	A+14
Splicing results	Normal
RNA-seq result	None
Site of splice aberration	None

<https://canvaruk.org/>

CanVar-UK: Diagnostic User forum

BRCA1 c.5207T>C p.(Val1736Ala)

[Download a variant report](#) [Click to search another variant](#)

GEN
INFO

Summary

In Silico Predictions

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Functional Splicing Analysis

CanVar-UK Diagnostic
Discussion Forum

Classification

— Proposed classification —

☐ Evaluation under a reduced penetrance model was considered for this variant

[Choose file](#) No file chosen

Extra relevant de-identified information (such as .doc, .pdf, .zip, .xls files) can be uploaded with your entry

Notes/evidence on the variant

Please select one of the options:

- ☐ 1. Save and email to all forum members: URGENT
- ☐ 2. Save and email to all forum members: STANDARD
- ☐ 3. Save and email only to forum members in the below discussion
- ☐ 4. Save only

POST

Cancel/Clear

Notes

#67136

Posted by: [Maia Le Dimeet \(UMR 1245\)](#) 2025-06-16 13:46:20

Proposed classification: Not provided

#66571

Posted by: [Alison Callaway \(Salisbury NHS Foundation Trust\)](#) 2024-05-22 16:54:59

Proposed classification: 5 - Pathogenic

Previously classified by CanVG in 2019. Have reviewed the classification locally in the context of a new patient and agree the variant is pathogenic. However, the combined LLR on the genetic epidemiology tab is -3.35/-10.53 (Easton LLR/ACMG LLR), which would equate to BP5_VS. We have discounted this piece of evidence as we believe there are issues with using LLR ratios / multifactorial analysis for assessment of benignity since co-occurrence in trans with pathogenic variants is possible and not embryonic lethal, apparent non-segregation can arise due to hypomorphic variants leading to later onset / reduced penetrance and also due cancer phenocopies.

#65660

Posted by: [Sophie Allen \(ICR\)](#) 2021-12-14 15:54:14

Attached file: [CLINVA-1.DOC](#)

Proposed classification: 5 - Pathogenic

CanVG discussion summary. Originally submitted 24/03/2019

#153

Posted by: [Tariq Mubtashid \(David Muscatelli MRC Foundation Trust\)](#) 2025-03-27 09:00:00

CanVar-UK: diagnostic-user forum

CanVar-UK

(NSW Health Pathology) commented on 2023-10-03 00:25 on MSH2 c.2635-3C>G.

Please share via the CanVar-UK platform any classifications, local data or comments about this variant using the black button below.

We have identified this variant in a Lynch Syndrome family with moderate segregation (Bayes factor of 14.65) PP1-mod: tumour characteristics meet PP4-supp; Absent from population; PM2-supp and in silico splicing analyses meet PP3; have performed RNA analysis and their data indicate variant results in skipping of exon 16, introducing a termination codon but they do not indicate whether of the normal and variant transcripts was done. Do you have additional information regarding individuals with this variant characteristics/segregation data? Thank you

Proposed classification: 3-Uncertain/review

[Click to view/comment](#)

Regards,
CanVar-UK

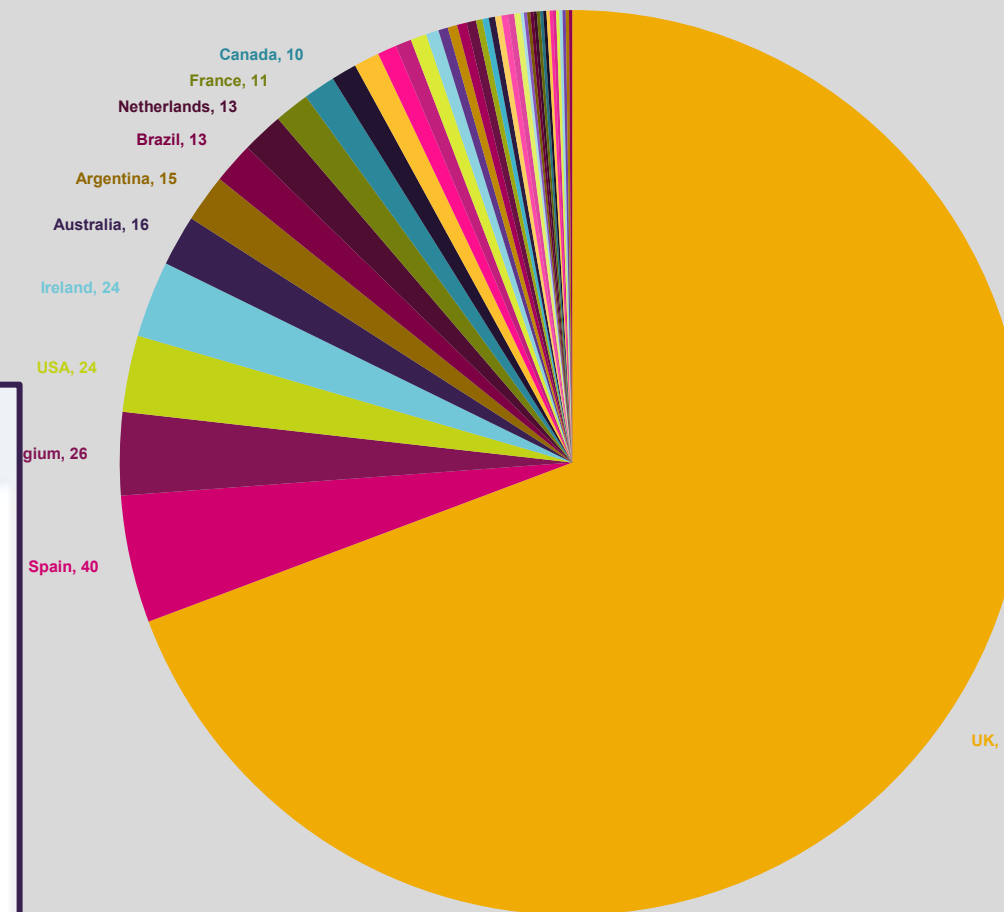
CanVar-UK

(Netherlands Cancer Institute) commented on 2023-10-03 08:50 on MSH2 c.2635-3C>G.

Please share via the CanVar-UK platform any classifications, local data or comments about this variant using the black button below.

Hi In The Netherlands we now consider this variant as pathogenic (class 5) based on the Invitae data (Invitae reports the result of mRNA testing: r.2635_*29353del, this effect is really strong and it is close to haploinsufficiency.), not in gnomAD non-cancer, co-segregation (2 dutch families), MSH-high, dMMR (MSH2/MSH6-), different MSH2 somatic second hits in independent tumors and dMMR in the ODMS functional assay (CIMRA assay not applicable for intronic variants). Best regards,

Proposed classification: 5-Pathogenic



879 members of diagnostic user forum, including 270 international (non-UK) members

Issues with ACMG v3.0 (2015) framework

1) Loose definitions/terminology

- Requirement for additional specification: ClinVar SVI/VCEPs, CanVIG-UK

2) Combinations of evidence items:

- Multiple data items non-orthogonal

3) Categorical, unscaled, empirical.

4) Universal vs disease-specific evidence items

- Variant Curation Expert Panels ('VCEPs') can vary in
 - How they define/interpret **universal** evidence elements
 - strictness for disease-specific evidence

Non-independent evidence items

Garrett et al, **Journal of Medical Genetics**, 2020

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Issues with ACMG v3.0 (2015) framework

1) Loose definitions/terminology

- Requirement for additional specification: ClinVar SVI/VCEPs, CanVIG-UK

2) Combinations of evidence items:

- Multiple data items non-orthogonal

3) Categorical, unscaled, empirical.

4) Universal vs disease-specific evidence items

- VCEPs variable in
 - defining **universal** evidence elements
 - strictness for disease-specific evidence

Categorical Framework

Moving to a scaled quantitative framework

- Categorical framework
 - Lacks flexibility for how evidence items are combined
 - Lacks flexibility in quantifying weight for evidence item (esp at higher end)

Table 5 Rules for combining criteria to classify sequence variants	
Pathogenic	(i) 1 Very strong (PVS1) AND (a) ≥1 Strong (PS1–PS4) OR (b) ≥2 Moderate (PM1–PM6) OR (c) 1 Moderate (PM1–PM6) and 1 supporting (PP1–PP5) OR (d) ≥2 Supporting (PP1–PP5) (ii) ≥2 Strong (PS1–PS4) OR (iii) 1 Strong (PS1–PS4) AND (a) ≥3 Moderate (PM1–PM6) OR (b) 2 Moderate (PM1–PM6) AND ≥2 Supporting (PP1–PP5) OR (c) 1 Moderate (PM1–PM6) AND ≥4 supporting (PP1–PP5)
Likely pathogenic	(i) 1 Very strong (PVS1) AND 1 moderate (PM1–PM6) OR (ii) 1 Strong (PS1–PS4) AND 1–2 moderate (PM1–PM6) OR (iii) 1 Strong (PS1–PS4) AND ≥2 supporting (PP1–PP5) OR (iv) ≥3 Moderate (PM1–PM6) OR (v) 2 Moderate (PM1–PM6) AND ≥2 supporting (PP1–PP5) OR (vi) 1 Moderate (PM1–PM6) AND ≥4 supporting (PP1–PP5)
Benign	(i) 1 Stand-alone (BA1) OR (ii) ≥2 Strong (BS1–BS4)
Likely benign	(i) 1 Strong (BS1–BS4) and 1 supporting (BP1–BP7) OR

Likelihood Ratio	Exponent points	Evidence Strength
LR towards pathogenicity		
		SUP
		MOD
		STRONG
		V/STRONG
LR towards Benignity		
		SUP
		STRONG

Tavtigian SV et al **Genetics in medicine** 2018
Garrett et al, **Journal of Medical Genetics**, 2020
Tavtigian SV et al **Hum Mut** 2021

Categorical Framework

Moving to a scaled quantitative framework

Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework

Sean V. Tavtigian, PhD¹, Marc S. Greenblatt, MD, PhD², Steven M. Harrison, PhD³, Robert L. Nussbaum, MD⁴, Snehit A. Prabhu, PhD⁵, Kenneth M. Boucher, PhD⁶ and Leslie G. Biesecker, MD⁷;

on behalf of the ClinGen Sequence Variant Interpretation Working Group (ClinGen SVI)

Purpose: We evaluated the American College of Medical Genetics and Genomics/Association for Molecular Pathology (ACMG/AMP) variant pathogenicity guidelines for internal consistency and compatibility with Bayesian statistical reasoning.

Methods: The ACMG/AMP criteria were translated into a naive Bayesian classifier, assuming four levels of evidence and exponentially scaled odds of pathogenicity. We tested this framework with a range of prior probabilities and odds of pathogenicity.

Results: We modeled the ACMG/AMP guidelines using biologically plausible assumptions. Most ACMG/AMP combining criteria were compatible. One ACMG/AMP likely pathogenic combination was mathematically equivalent to pathogenic and one ACMG/AMP pathogenic combination was actually likely pathogenic. We modeled combinations that include evidence for and against pathogenicity, showing that our approach scored some combinations as pathogenic or likely pathogenic that ACMG/AMP would

designate as variant of uncertain significance (VUS).

Conclusion: By transforming the ACMG/AMP guidelines into a Bayesian framework, we provide a mathematical foundation for what was a qualitative heuristic. Only 2 of the 18 existing ACMG/AMP evidence combinations were mathematically inconsistent with the overall framework. Mixed combinations of pathogenic and benign evidence could yield a likely pathogenic, likely benign, or VUS result. This quantitative framework validates the approach adopted by the ACMG/AMP, provides opportunities to further refine evidence categories and combining rules, and supports efforts to automate components of variant pathogenicity assessments.

Genet Med advance online publication 4 January 2018

Key Words: Bayesian framework; medical genetics; unclassified variants; variant classification; variants of uncertain significance

Likelihood Ratio	Exponent points	Evidence Strength
LR towards pathogenicity		
2.1		SUP
4.3		MOD
18.7		STRONG
350.4		V/STRONG
LR towards Benignity		
0.48		SUP
0.05		STRONG

Tavtigian SV et al **Genetics in medicine** 2018

Garrett et al, **Journal of Medical Genetics**, 2020

Tavtigian SV et al **Hum Mut** 2021

Categorical Framework

Moving to a scaled quantitative framework

Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework

Position statement

OPEN ACCESS

POSITION STATEMENT

Combining evidence for and against pathogenicity for variants in cancer susceptibility genes: CanVIG-UK consensus recommendations

Alice Garrett ¹, Miranda Duijck ¹, Sean V. Tavtigian ^{1,2}, Steven M. Harrison ³, Kenneth M. Boucher ^{2,4}, Leslie G. Biesecker ⁵

Hum Mutat. 2020 October ; 41(10): 1734–1737. doi:10.1002/humu.24088.

Fitting a naturally scaled point system to the ACMG/AMP variant classification guidelines

Sean V. Tavtigian^{1,2}, Steven M. Harrison³, Kenneth M. Boucher^{2,4}, Leslie G. Biesecker⁵

Evidence strength	Odds of pathogenicity	Exponent score
Very Strong (VS)	350	2.08 ^a 8
Benign	0.01 (2.08 ⁻⁶)	≤-6 0.001*

*The posterior probability attained with an exponent sum of -6 has been rounded down to 0.001, consistent with the UK Association for Clinical Genomic Science Best Practice Guidelines for Variant Classification in Rare Disease 2020.⁴

Tavtigian SV et al **Genetics in medicine** 2018

Garrett et al, **Journal of Medical Genetics**, 2020

Tavtigian SV et al **Hum Mut** 2021

Likelihood Ratio	Exponent points	Evidence Strength
LR towards pathogenicity		
2.1	1	SUP
4.3	2	MOD
18.7	4	STRONG
350.4	8	V/STRONG
LR towards Benignity		
0.48	-1	SUP
0.05	-4	STRONG

Categorical Framework

Moving to a scaled quantitative framework

- ACMG framework update 2026 (v4.0)
- Move to points-based system

Likelihood Ratio	Exponent points	Evidence Strength
LR towards pathogenicity		
2.1	1	SUP
4.3	2	MOD
9.0	3	
18.7	4	STRONG
38.9	5	
81.0	6	
168.4	7	
350.4	8	V/STRONG
LR towards Benignity		
0.48	-1	SUP
0.23	-2	
0.11	-3	
0.05	-4	STRONG

Tavtigian SV et al **Genetics in medicine** 2018
Garrett et al, **Journal of Medical Genetics**, 2020
Tavtigian SV et al **Hum Mut** 2021

Issues with ACMG v3.0 (2015) framework

1) Loose definitions/terminology

- Requirement for additional specification: ClinVar SVI/VCEPs, CanVIG-UK

2) Combinations of evidence items:

- Multiple data items non-orthogonal

3) Categorical, unscaled, empirical.

4) Universal vs disease/gene-specific evidence items

- Variant Curation Expert Panels ('VCEPs') can vary in
 - How they define/interpret **universal** evidence elements
 - Strictness/permmissiveness for disease-specific evidence

Approaches/frameworks for variant classification

- 1) **Reductive**: Variant detected in disease→variant pathogenic
- 2) **Multifactorial**: statistical quantitation (centralised)
- 3) **ACMG v3.0 2015**: categorical framework (decentralised)
- 4) **ACMG v4.0 2026**: quantitative **AND** categorical framework (decentralised)

Complexities/challenges: Classification Frameworks

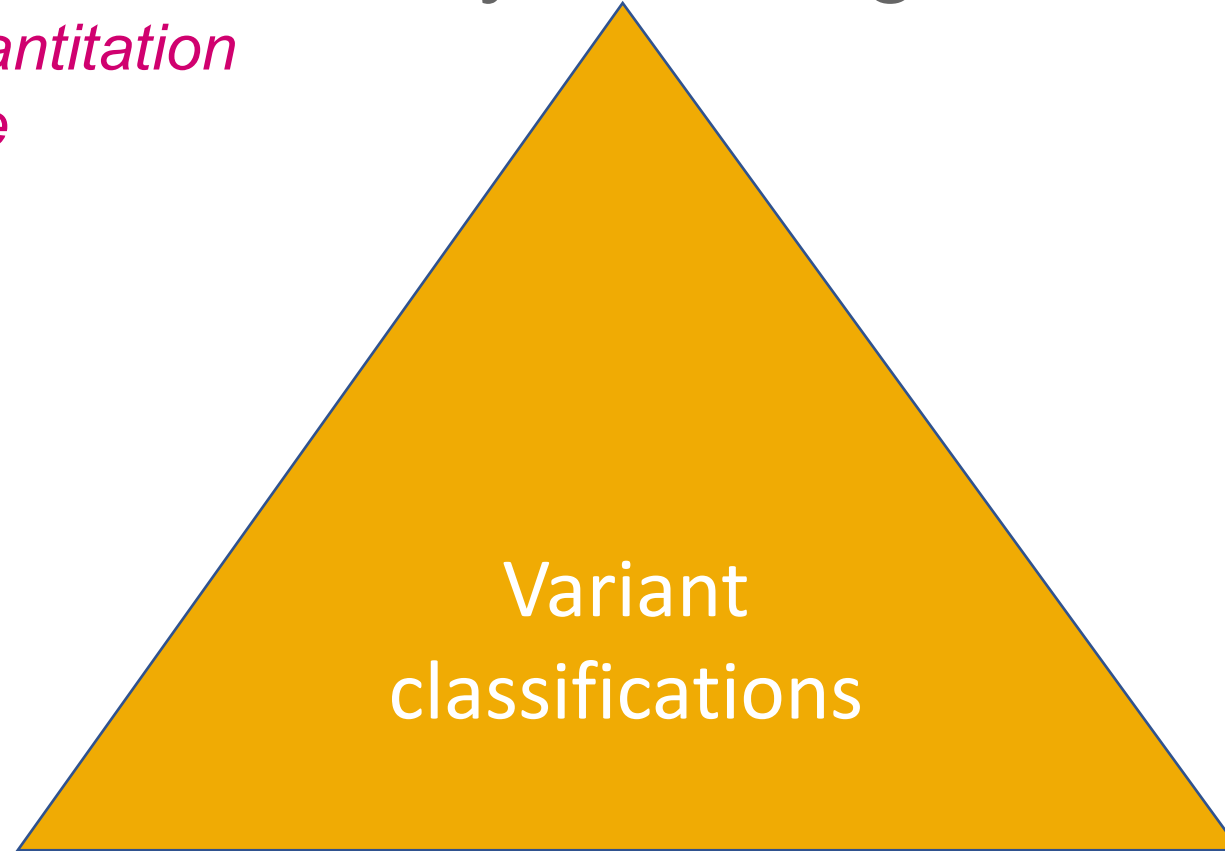
- ACMG Framework v3.0 under continuous evolution/improvement since 2015
 - Address **non-independence** of evidence elements
 - Transition from **categorical to quantitative** system
 - Address **universal rules versus gene-specific rules; equivalence of applications**
- Complexity of Framework:
 - 27 evidence items for evaluation
 - Manual assembly of data: multisource, time-consuming (should be re-evaluated each time variant is seen)
- Tension of “devolved” model with application by jobbing diagnostic–lab scientists versus “experts-only” (centralised, VCEP) model
- Considerable inconsistency in weighting, tallying and final classifications
- Down-classification highly problematic
- Value of national network: CanVIG-UK meetings, CanVar-UK discussion forum
- ACMG Framework v4.0 : large-step transition from v3.0. Sizeable impact on labs

Themes

1. Clinical Genetics
2. Phenotypes and ascertainment
3. Variant-specific risks/genotype-phenotype
4. Evolution of Variant Interpretation Frameworks
5. MAVES, clinical variant interpretation and opportunities forward

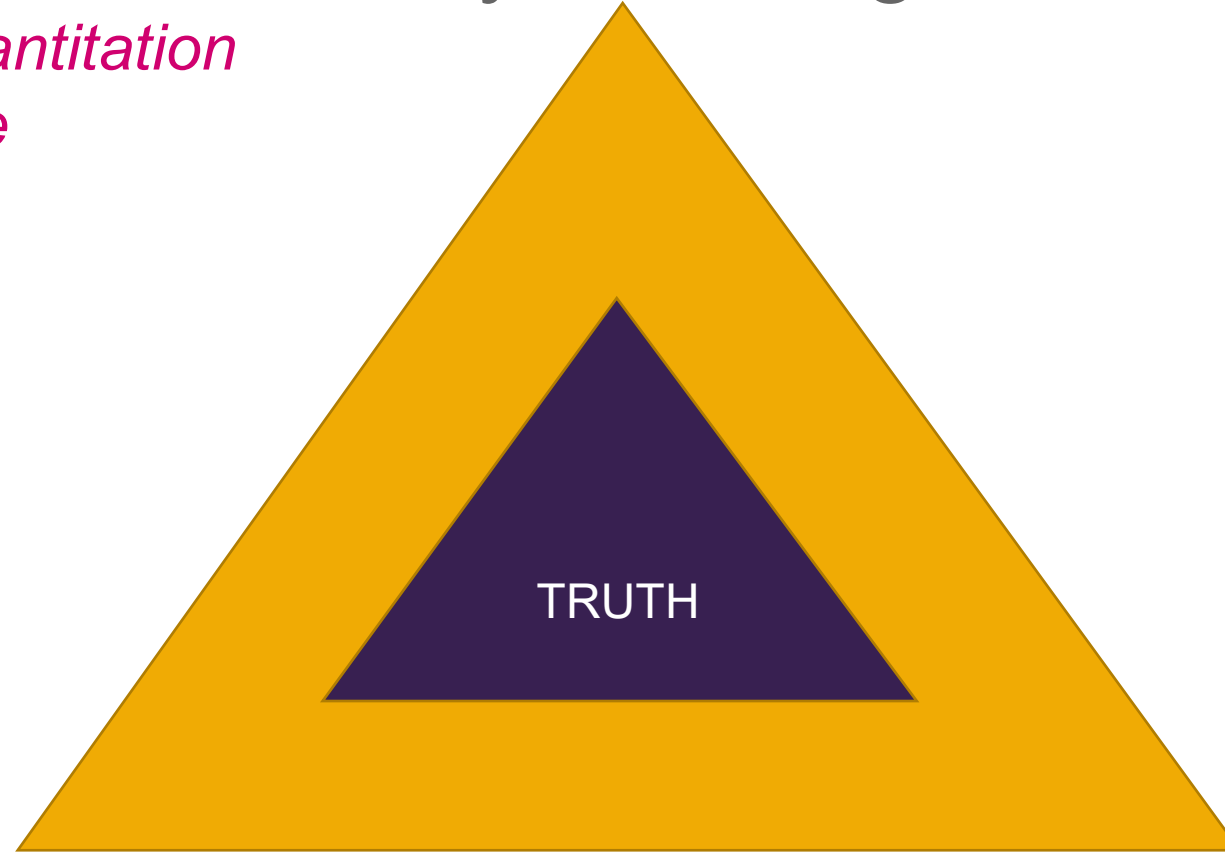
Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*



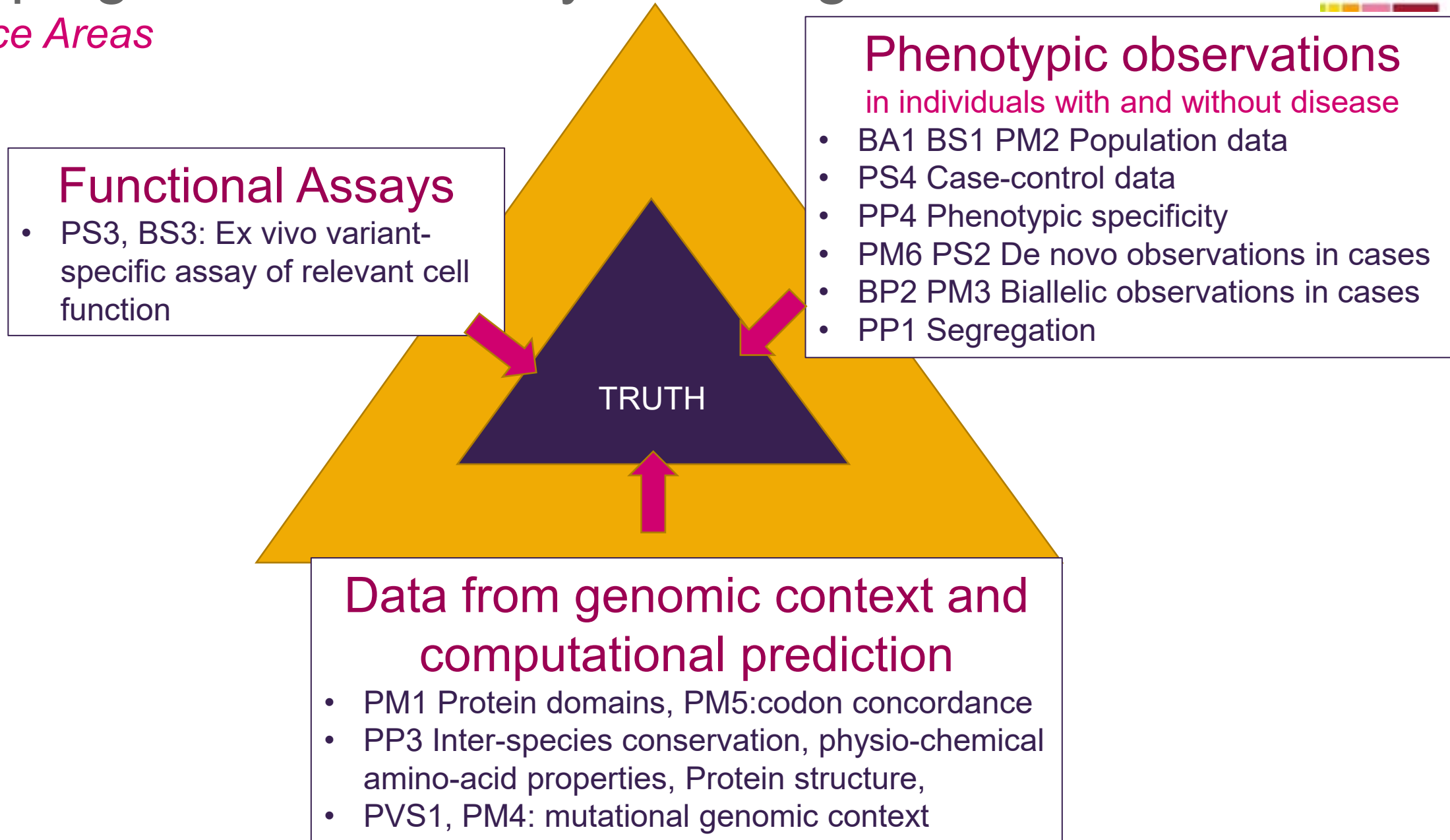
Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*



Developing best community variant gold-truth

3 Evidence Areas



Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*

Functional Assays

- PS3, BS3: Ex vivo variant-specific assay of relevant cell function

Phenotypic observations in individuals with and without disease

- BA1 BS1 PM2 Population data
- PS4 Case-control data
- PP4 Phenotypic specificity
- PM6 PS2 De novo observations in cases
- BP2 PM3 Biallelic observations in cases
- PP1 Segregation

TRUTH

Data from genomic context and computational prediction

- PM1 Protein domains, PM5:codon concordance
- PP3 Inter-species conservation, physio-chemical amino-acid properties, Protein structure,
- PVS1, PM4: mutational genomic context

Cross-
validating with
other data from
other sources

Developing best community variant gold-truth

Handful of
variants.
Limited
reproducibility

Best quantitation
evidence

Functional Assays

- PS3, BS3: Ex vivo variant-specific assay of relevant cell function

Phenotypic observations

in individuals with and without disease

- BA1 BS1 PM2 Population data
- PS4 Case-control data
- PP4 Phenotypic specificity
- PM6 PS2 De novo observations in cases
- BP2 PM3 Biallelic observations in cases
- PP1 Segregation

Case data
typically
sparse or
non-existent

TRUTH

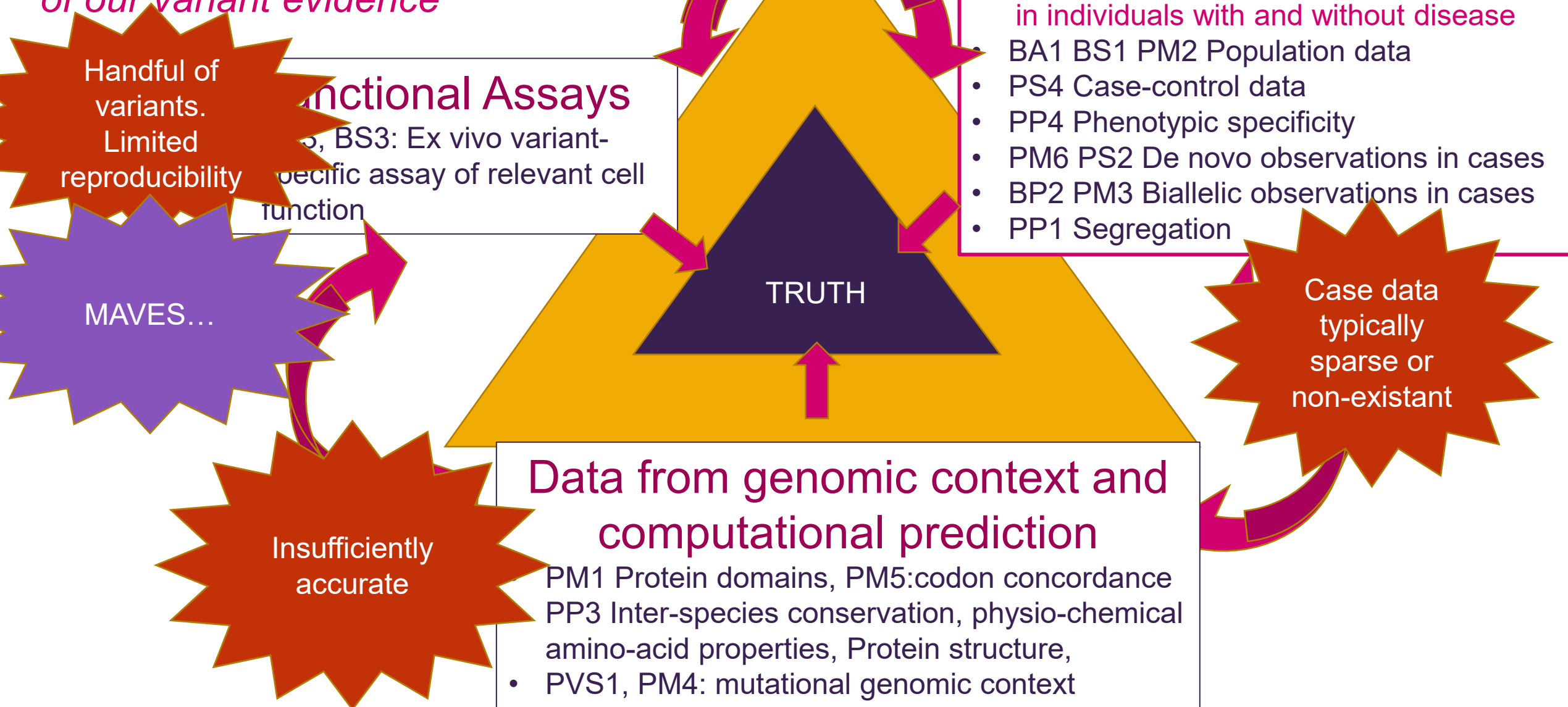
Data from genomic context and computational prediction

- PM1 Protein domains, PM5:codon concordance
- PP3 Inter-species conservation, physio-chemical amino-acid properties, Protein structure,
- PVS1, PM4: mutational genomic context

Insufficiently
accurate

Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*



Developing best community variant gold-truth

Improving information from phenotypic observation

Functional Assays

- PS3, BS3: Ex vivo variant-specific assay of relevant cell function

Phenotypic observations

in individuals with and without disease

- BA1 BS1 PM2 Population data
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- BP2 PM3 Biallelic observations in cases
- PP1 Segregation

TRUTH

MAVES...

Insufficiently accurate

Data from genomic context and computational prediction

- PM1 Protein domains, PM5:codon concordance
- PP3 Inter-species conservation, physio-chemical amino-acid properties, Protein structure,
- PVS1, PM4: mutational genomic context

Developing best community variant gold-truth

Improving information from phenotypic observation

Functional Assays

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MAVES...

Phenotypic observations

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TRUTH

Data from genomic context and computational prediction

- PM1 Protein domains, PM5:codon concordance
- PP3 Inter-species conservation, physio-chemical amino-acid properties, Protein structure,
- PVS1, PM4: mutational genomic context

Advancing how clinical evidence is apportioned: PS4: case control data

	Benign		Pathogenic			
	Strong	Supporting	Supporting	Moderate	Strong	Very strong
Population data	MAF is too high for disorder BA1/BS1 OR observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data →		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in trans with a dominant variant BP2 Observed in cis with a pathogenic variant BP2		For recessive disorders, detected in trans with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

PS4: The prevalence of the variant in affected individuals is significantly increased compared with the prevalence in controls

Relative risk or OR, as obtained from case–control studies, is >5.0, and the confidence interval around the estimate of relative risk or OR does not include 1.0.

1. Applicable only at **strong**
2. Inflexible criteria for applicability
3. Not applicable as evidence of **benignity**

Not adapted for quantitative framework by ClinGen SVI or any VCEP

	Cases	Controls
Variant ⁺	a	b
Variant ⁻	c	d

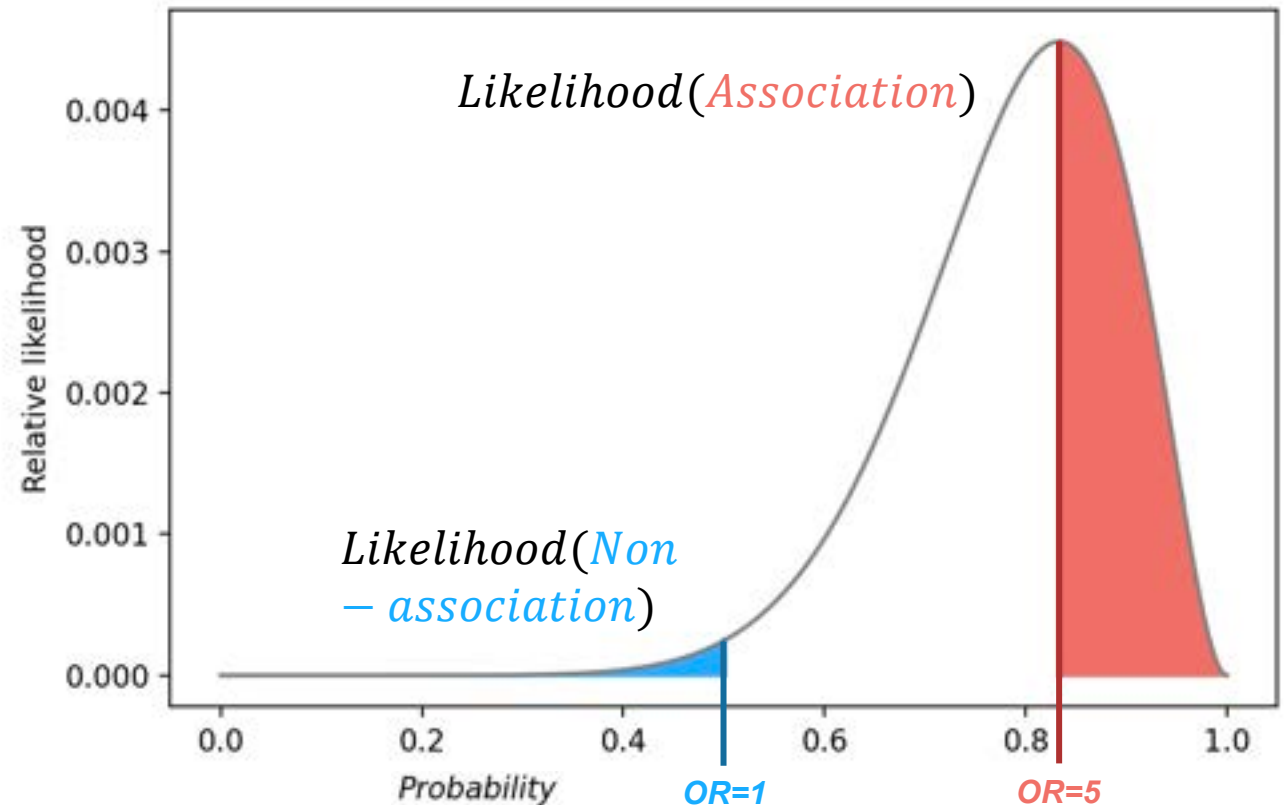
$$\text{Observed odds ratio} = \frac{a/b}{c/d}$$

Hypothesis of **association**

OR $\geq$ target odds of **association** (e.g. 5)
→ indicative of **pathogenicity**

Hypothesis of **non-association**

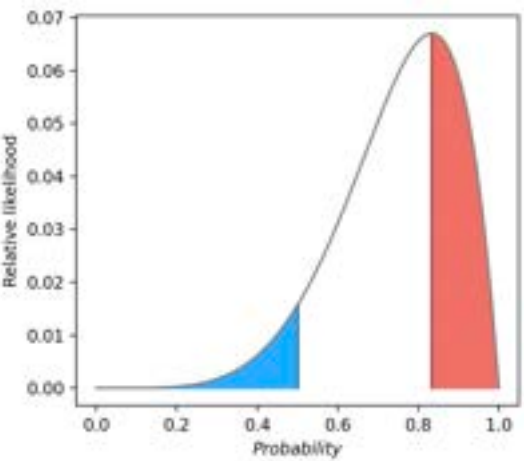
OR $\geq$ target odds of **non-association** (e.g. 5)
→ indicative of **benignity**



$$LR = \frac{\text{Likelihood}(\text{Pathogenic})}{\text{Likelihood}(\text{Benign})}$$

	Cases	Controls
V ⁺	5	1
V ⁻	10,000	10,000

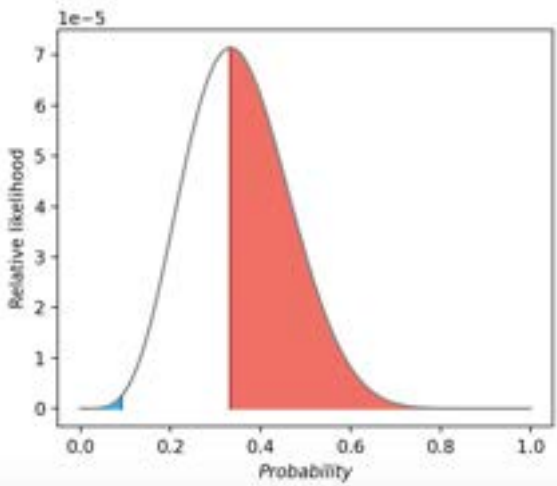
OR = 5.0 (0.58-42.80)



OR ≥ 5
OR ≤ 1

	Cases	Controls
V ⁺	5	10
V ⁻	10,000	100,000

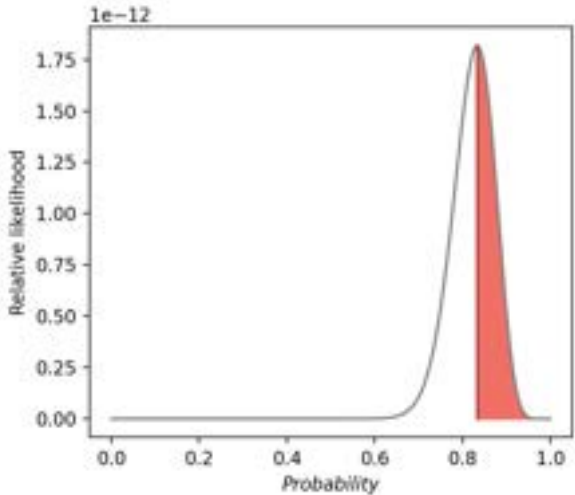
OR = 5.0 (1.71-14.63)



OR ≥ 5
OR ≤ 1

	Cases	Controls
V ⁺	50	10
V ⁻	100,000	100,000

OR = 5.0 (2.54-9.86)



PS4 (ACMG v3)

No evidence

PS4-LRCalc

LR = 5.28

EPs = 2.27

Moderate evidence

PS4 (ACMG v3)

EPs = 4
Strong evidence

PS4-LRCalc

LR = 270.44

EPs = 7.65

Strong evidence

PS4 (ACMG v3)

EPs = 4
Strong evidence

PS4-LRCalc

LR = 9.13x10⁶

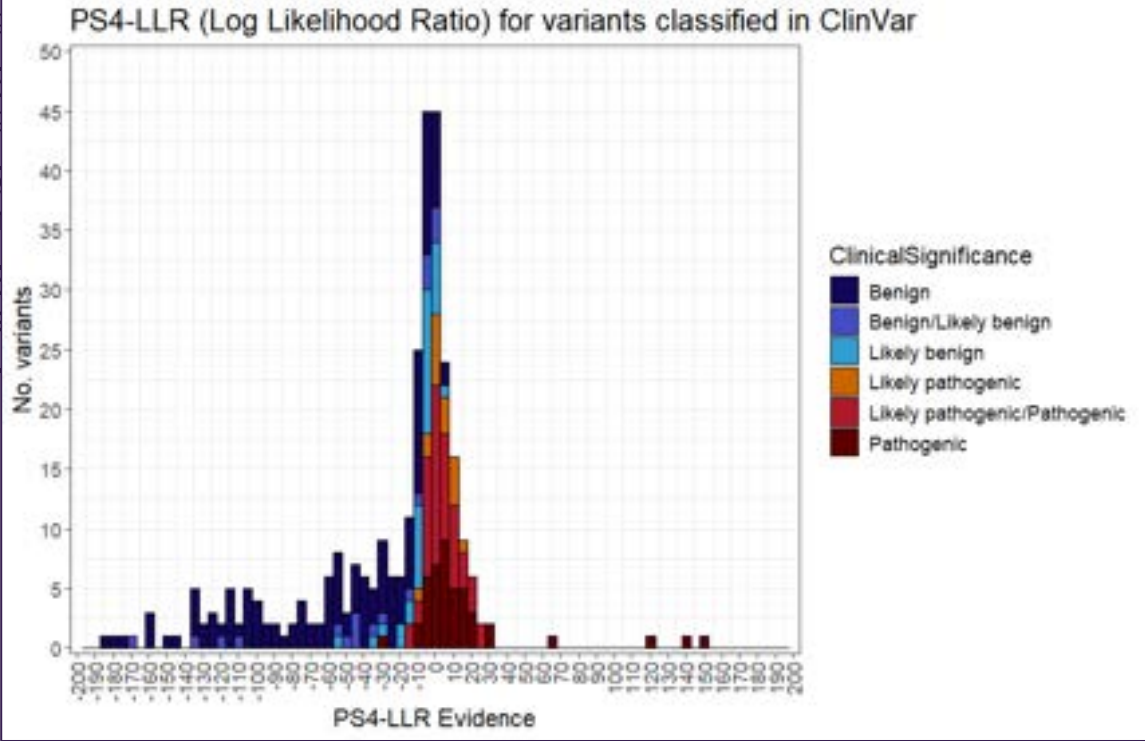
EPs = 21.88

Very strong evidence

Applying PS4-LRC to breast cancer data

Dataset	Genes	Ascertainment	Cases	Control	
UK Biobank	ATM, BRCA1, BRCA2, CHEK2, PALB2	Unselected	18,477	419,373	
CARRIERS	ATM, BRCA1, BRCA2, CHEK2, PALB2	Unselected	32,247	32,544	
BRIDGES	ATM, BRCA1, BRCA2, CHEK2, PALB2	Unselected	42,062	44,035	
Ambry	ATM, BRCA1, BRCA2, CHEK2, PALB2	Clinical	187,642	NA	
NDRS-UK	BRCA1, BRCA2	Clinical	44,917	NA	UK
gnomAD v4.1.0	ATM, BRCA1, BRCA2, CHEK2, PALB2	Population	NA	175,054	Global
Total			325,345	671,006	

	TOTAL VARIANTS	Any pathogenic evidence (% total)	No evidence (% total)	Any benign evidence (% total)	PS4-LLR Evidence towards pathogenicity (LLR≥1)				PS4-LLR Evidence towards benignity (LLR≤-1)			
					VSTR	STR	MOD	SUP	SUP	MOD	STR	VSTR
BRCA1	899	108 (12)	75 (8)	716 (80)	21	23	23	41	23	62	182	449
BRCA2	1722	175 (10)	136 (8)	1411 (82)	26	37	41	71	37	204	397	773
PALB2	559	84 (15)	49 (9)	426 (76)	4	10	33	37	16	90	104	216
ATM	1371	363 (27)	11 (1)	1327 (97)								
CHEK2	380	143 (38)	3 (1)	374 (98)								
TOTAL	4931	873 (17)	4 (0)	4923 (99)								



Allen et al, 2025 manuscript in draft

Advancing how clinical evidence is apportioned: PP4: phenotypic specificity

Benign		Pathogenic				
	Strong	Supporting	Supporting	Moderate	Strong	Very strong
Population data	MAF is too high for disorder BA1/BS1 OR observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
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Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

PP4: Patient's phenotype or family history is highly specific for disease

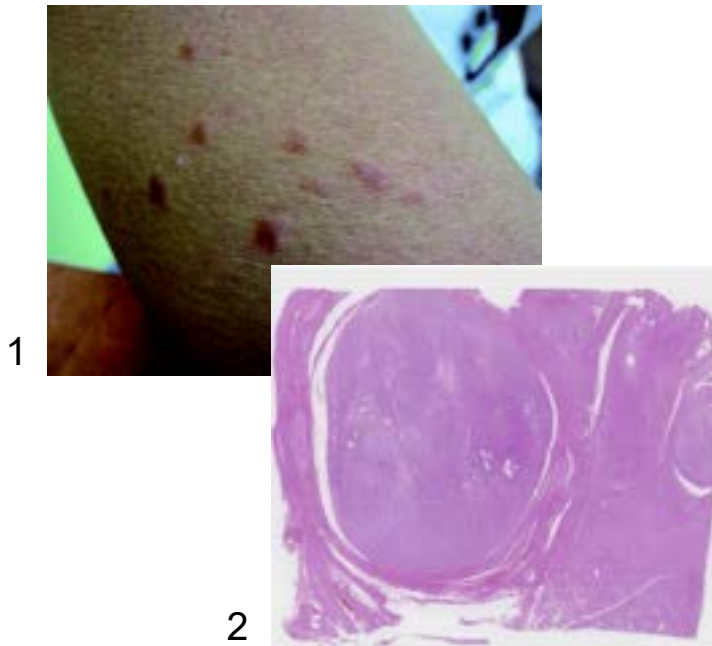
1. Applicable at **supporting**
2. Weighting deliberately conservative (as phenotype was historically over-weighted)
3. Variability in how weighted clinically and by VCEPs

Not adapted for quantitative framework by ClinGen SVI

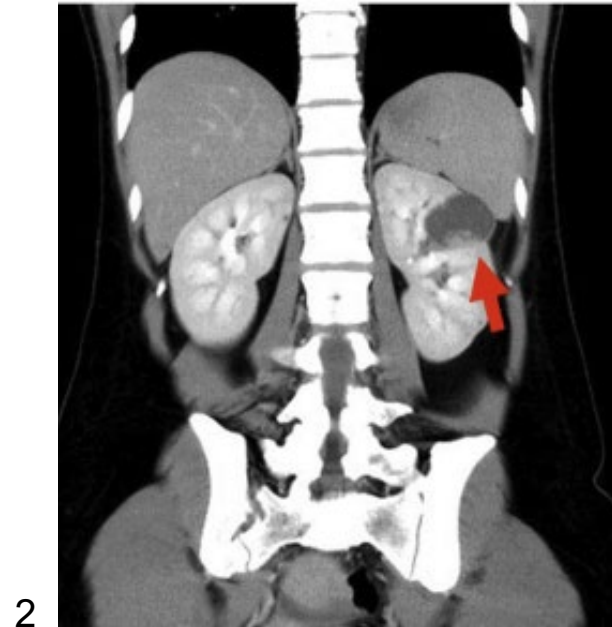
Quantifying specificity for very rare variants in fumarate hydratase gene (*FH*)

2 contexts of clinical testing: (1) Hereditary Leiomyomatosis and Renal Cell Cancer (HLRCC) (2) renal cancer

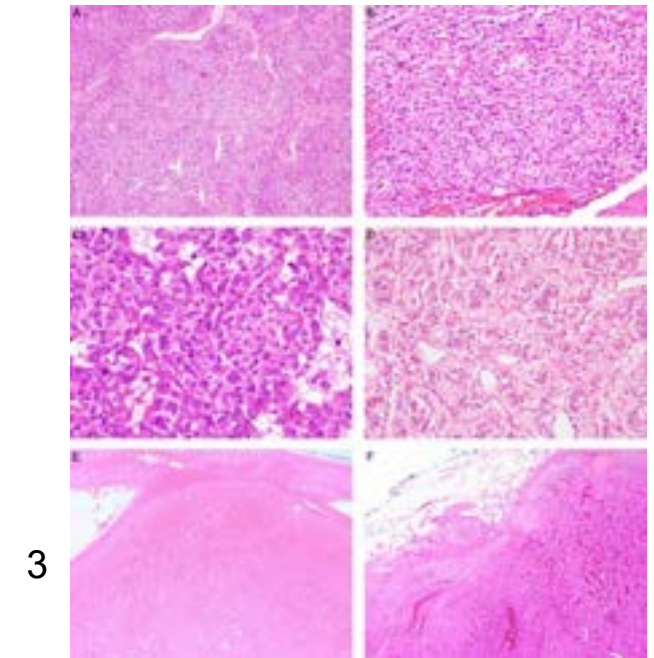
Cutaneous and Uterine leiomyomas



Renal cell cancer



Phaeochromocytoma



1: Varol et al., 2006. doi:10.1111/j.1440-0960.2006.00294.x

2: Yonamine et al., 2020. doi:10.1016/j.eucr.2020.101141

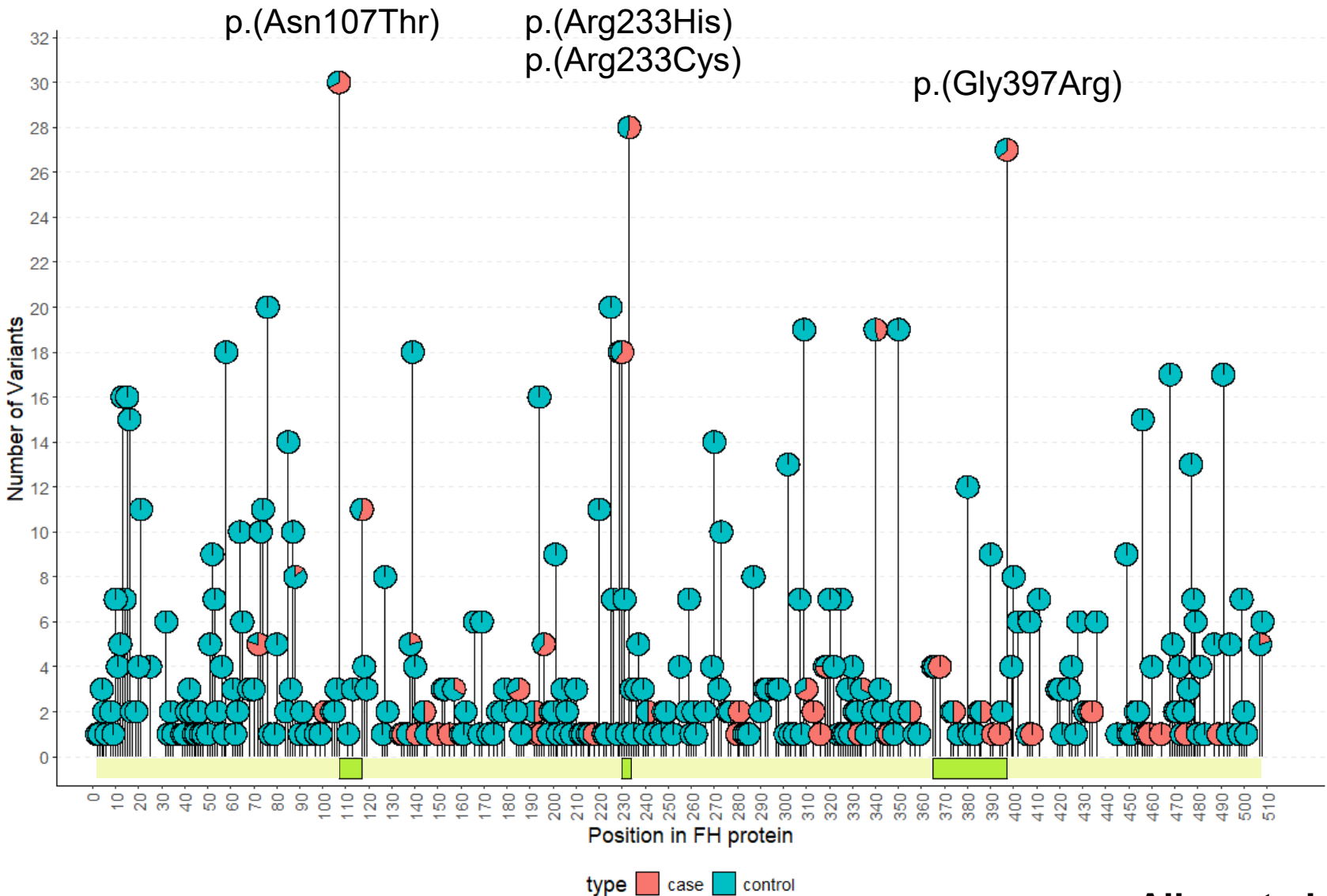
Very Rare (missense) Variant Likelihood Ratio

2 contexts of clinical testing: (1) Hereditary Leiomyomatosis and Renal Cell Cancer (HLRCC) (2) renal cancer

Use MTAF to define a very rare variant

		Cases with variant	Controls with variant (/562,290 controls)	Likelihood Ratio (95% Confidence Intervals)	Evidence Points
HLRCC (/387 cases)	Rare Truncating Variants	79	43	2669.39 (1843.44-3881.25)	+10.77
	Rare Missense Variants	146	988	214.71 (185.01-246.87)	+7.33
Renal (/1780 cases)	Rare Truncating Variants	13	43	95.50 (48.93-183.0)	+6.23
	Rare Missense Variants	18	988	5.76 (3.51-9.30)	+2.39

Clustering of missense variants in HLRCC cases versus population controls



	LR (95% Confidence Intervals)	ACMG Points
Total pan-gene	214.71 185.01-246.87	+7.33
Inside clustered regions	1225.92 (918.63-1631.95)	+9.71
Outside clustered region	105.88 (82.97-133.65)	+6.37

Quantifying specificity for very rare variants in SDHB/SDHD

Genetics in Medicine (2022) 24, 41–50



Genetics
Medicine

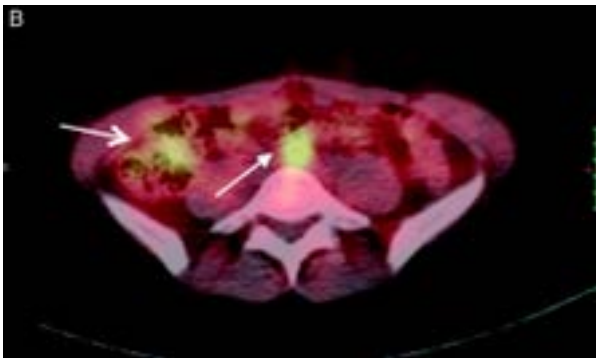
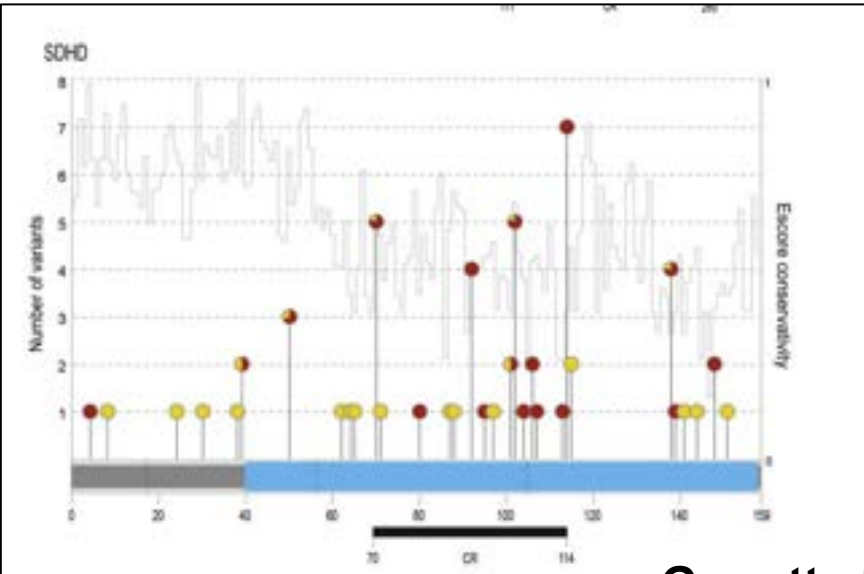
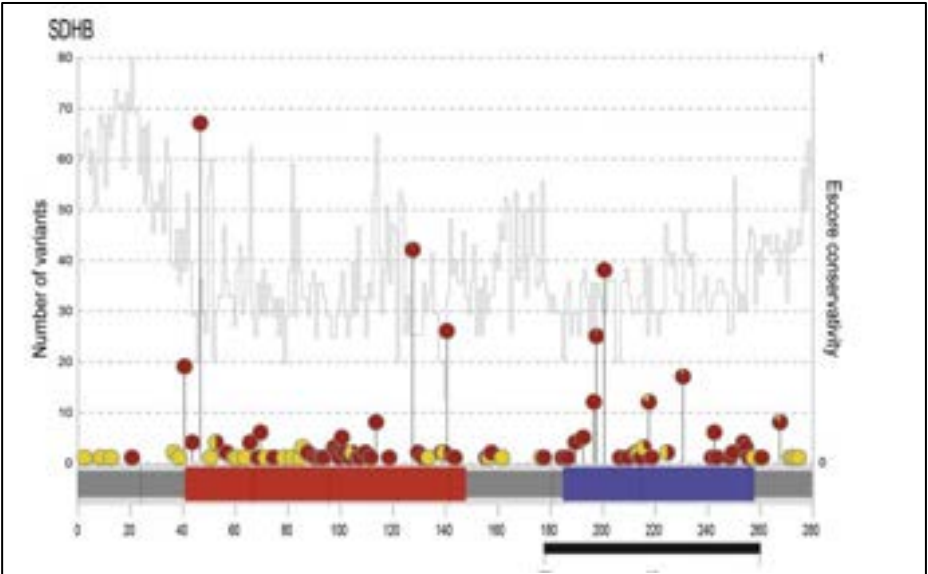
www.geneticsinmedicine.com

ARTICLE

Quantifying evidence toward pathogenicity for rare phenotypes: The case of succinate dehydrogenase genes, *SDHB* and *SDHD*

Alice Garrett¹, Chey Loveday¹, Laura King¹, Samantha Butler², Rachel Robinson³, Carrie Horton⁴, Amal Yussuf⁵, Subin Choi¹, Beth Torr¹, Miranda Durkie⁶, George J. Burghel⁶, James Drummond¹, Ian Berry⁵, Andrew Wallace⁶, Alison Callaway⁶, Diana Eccles^{6,10}, Marc Tischkowitz^{11,12}, Katrina Tatton-Brown^{13,14}, Katie Snape^{13,14}, Terri McVeigh¹⁵, Louise Izatt¹⁶, Emma R. Woodward^{17,18}, Nelly Burnichon^{19,20}, Anne-Paule Gimenez-Roqueplo^{19,20}, Francesco Mazzarotto^{21,24}, Nicola Whiffin^{21,22}, James Ware^{23,24}, Helen Hanson^{1,18}, Tina Pesaran⁴, Holly LaDuca⁴, Alexandre Buffet^{29,30}, Eamonn R. Maher³¹, Clare Turnbull^{1,15,4}; On behalf of Cancer Variant Interpretation Group UK (CanVIG-UK)

		VRMVs (all)			VRMVs (founders excluded)		
		PCC/PGL	Population controls	Positive Likelihood Ratio	PCC/PGL	Population controls	Positive Likelihood Ratio
SDHB	VRMV present	366	39	76.2 (54.8-105.9)	137	38	30.4 (21.2-43.5)
	VRMV absent	5,962	51,338		5,962	51,338	
	Total	6,328	51,377		6,099	51,376	
SDHD	VRMV present	37	22	14.8 (8.7-25.0)	37	22	14.8 (8.7-25.0)
	VRMV absent	5,810	51,355		5,810	51,355	
	Total	5,847	51,377		5,847	51,377	



Advancing how clinical evidence is apportioned: Benign classifications

Paucity of clinically-classified benign variants

Gene	ClinVar review status	# ClinVar-classified variants													
		P/LP							B/LB						
		All variants			Missense variants			% missense	All variants			Missense variants			% missense
		P	LP	P+LP	P	LP	P+LP		B	LB	B+LB	B	LB	B+LB	
Inherited breast cancer susceptibility															
BRCA1	≥2*	2641	31	2672	90	18	108	4.0%	746	1028	1774	152	31	183	10.3%
	≥1*	3358	175	3533	109	63	172	4.9%	789	2384	3173	160	142	302	9.5%
BRCA2	≥2*	3310	25	3335	41	7	48	1.4%	790	1761	2551	153	48	201	7.9%
	≥1*	4488	260	4748	45	24	69	1.5%	821	3572	4393	157	163	320	7.3%
PALB2	≥2*	624	39	663	1	0	1	0.2%	87	502	589	11	0	11	1.9%
	≥1*	1098	121	1219	1	1	2	0.2%	114	1165	1279	12	13	25	2.0%
ATM	≥2*	1057	131	1188	24	15	39	3.3%	244	1601	1845	25	0	25	1.4%
	≥1*	2270	469	2739	29	30	59	2.2%	374	4136	4510	31	20	51	1.1%
CHEK2	≥2*	287	68	355	3	1	4	1.1%	36	290	326	1	0	1	0.3%
	≥1*	658	187	845	3	3	6	0.7%	72	799	871	1	4	5	0.6%
Inherited ovarian cancer susceptibility															
RAD51C	≥2*	104	19	123	5	2	7	5.7%	26	193	219	2	0	2	0.9%
	≥1*	181	61	242	5	2	7	2.9%	35	486	521	3	1	4	0.8%
RAD51D	≥2*	49	22	71	0	0	0	0%	25	198	223	3	0	3	1.3%
	≥1*	119	70	189	0	0	0	0%	40	510	550	3	1	4	0.7%
BRIP1	≥2*	318	64	382	2	1	3	0.8%	69	541	610	5	0	5	0.8%
	≥1*	613	172	785	2	4	6	0.8%	110	1339	1449	5	18	24	1.7%

Rowlands et al, Am J Hum Genet 2025 112(10)

Cross validating between:

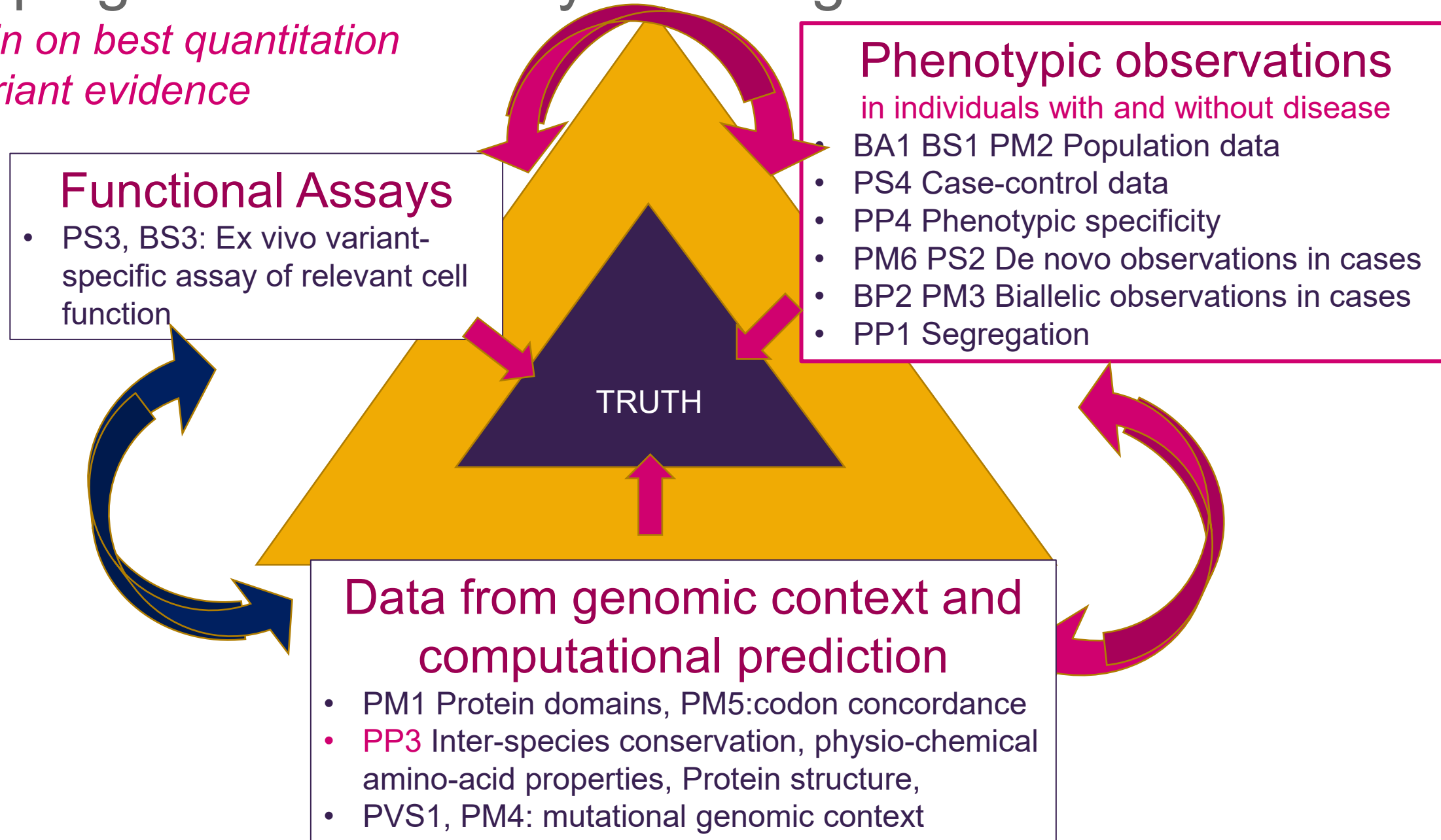
Clinical data

Functional data

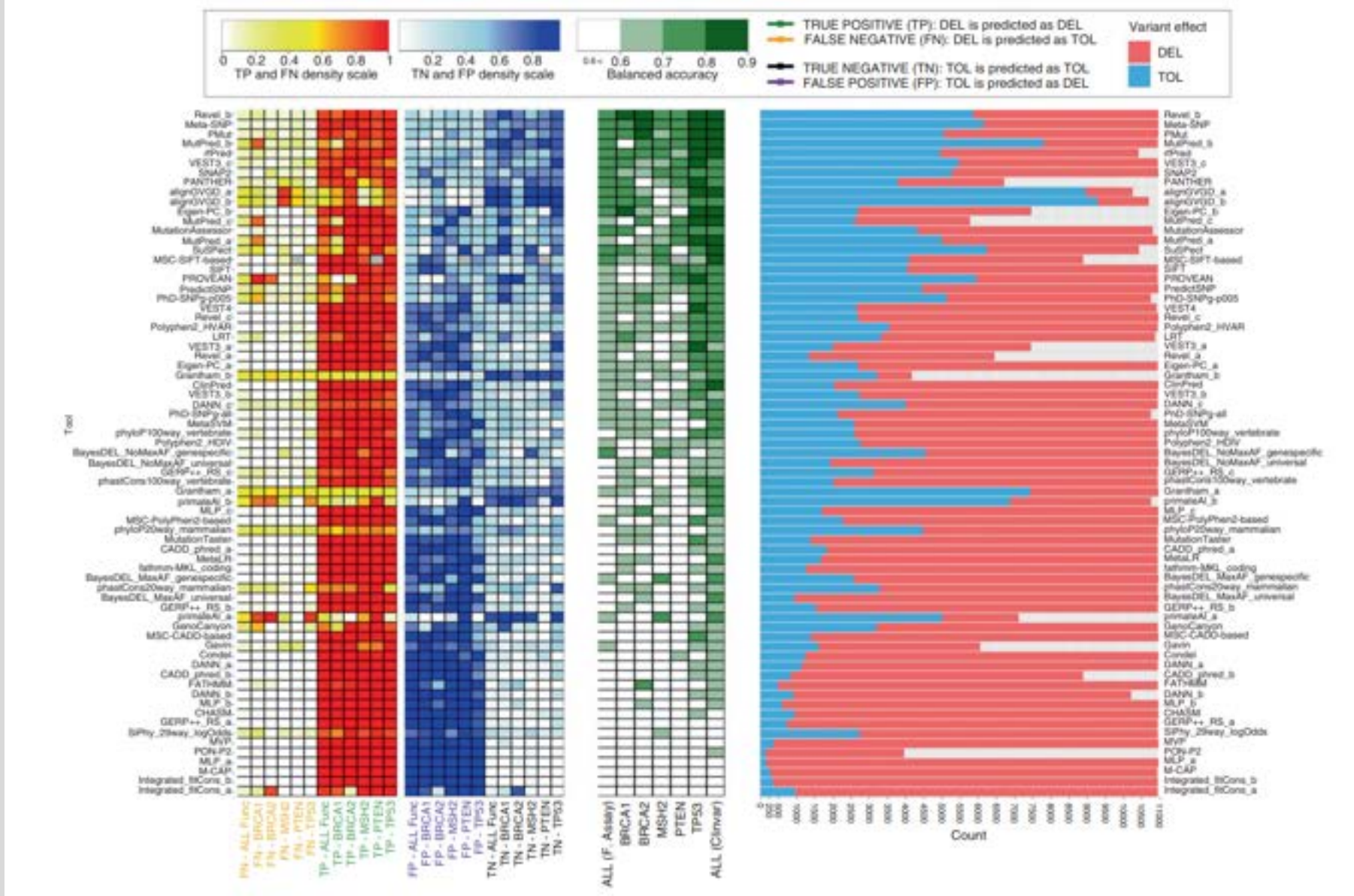
Predictive (in silico) data

Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*



Quantifying and cross-calibrating evidence items



- 44 in silico tools
- Truth-sets
 - 5 MAVEs
 - Individually
 - Combined
 - ClinVar
- Multiple metrics of tool performance
 - Balanced accuracy (BA)
 - Specific recommended thresholds
- Top performers
 - REVEL (0.7)
 - Meta-SNP (0.7)

Quantifying and cross-calibrating evidence items

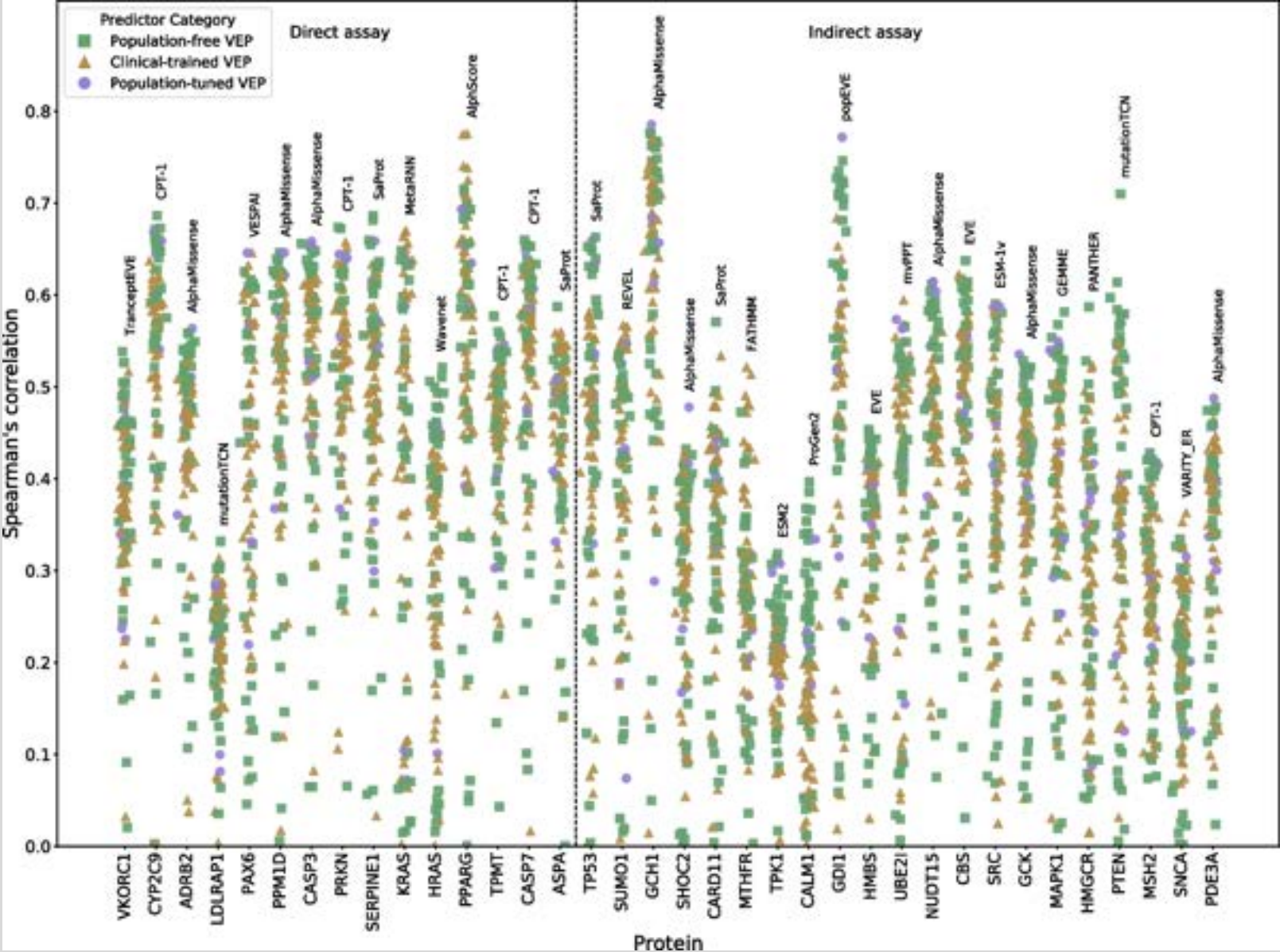
Functional assays↔ PP3: In silico tools

Table 2. Positive likelihood ratios (PLR) and negative likelihood ratios (NLR) for pathogenicity for REVEL and Meta-SNP predictions using binary thresholds and score bands, examining the mean of results for each gene for *BRCA1*, *BRCA2*, *MSH2*, *PTEN*, and *TP53* and the unweighted result for all variants (see also Supplementary Tables 10 and 11).

		REVEL				Meta-SNP			
		Mean		All		Mean		All	
		PLR	NLR	PLR	NLR	PLR	NLR	PLR	NLR
Binary	0.9	45.9 (9.63–302.)	1.98 (1.25–5.67)	5.01 (4.59–5.48)	1.69 (1.54–1.84)	32.9 (3.08–470)	1.01 (0.085–26.2)	49.2 (14.9–162)	1.01 (0.308–3.36)
	0.8	6.24 (4.83–8.22)	4.05 (3.37–4.98)	3.15 (3.00–3.31)	3.02 (2.87–3.17)	23.5 (8.69–203)	1.40 (0.773–5.99)	15.2 (12.7–18.2)	1.38 (1.15–1.65)
	0.7	3.13 (2.75–3.58)	7.20 (6.27–8.33)	2.33 (2.26–2.41)	5.39 (5.21–5.57)	6.80 (5.27–9.09)	3.64 (2.87–4.77)	5.68 (5.33–6.06)	3.43 (3.22–3.66)
	0.5	1.49 (1.40–1.59)	19.9 (18.7–21.2)	1.37 (1.35–1.39)	12.4 (12.2–12.6)	2.79 (2.49–3.14)	9.98 (8.81–11.3)	2.75 (2.66–2.85)	8.62 (8.32–8.92)
	0.4	1.18 (1.14–1.23)	19.0 (18.2–19.7)	1.15 (1.13–1.16)	10.6 (10.4–10.7)	2.02 (1.86–2.20)	15.4 (14.1–16.8)	1.88 (1.84–1.93)	14.3 (14.0–14.7)
	0.1	1.00 (1.00–1.00)	1.87 (1.87–1.88)	1.00 (1.00–1.00)	16.7 (16.7–16.7)	1.05 (1.03–1.07)	25.1 (24.6–25.6)	1.02 (1.02–1.03)	41.7 (41.5–41.9)
Band	0.9–1.0	102 (17.9–701)	Compared to REVEL < 0.7	6.19 (5.74–6.67)	Compared to REVEL < 0.7	122. (10.8–1,799)	Compared to Meta-SNP < 0.7	157 (47.9–518)	Compared to Meta-SNP < 0.7
	0.8–1.0	6.74 (5.24–8.82)		3.14 (3.00–3.28)		42.9 (14.4–406.)		24.9 (21.0–29.5)	
	0.8–0.9	6.77 (5.07–9.15)		3.77 (3.50–4.06)		42.7 (14.2–406.)		24.6 (20.7–29.2)	
	0.7–0.8	3.07 (2.23–4.28)		2.93 (2.62–3.26)		6.64 (5.01–9.07)		5.62 (5.20–6.07)	
	0.5–0.7	Compared to REVEL > 0.7	7.44 (6.64–8.38)	Compared to REVEL > 0.7	4.95 (4.81–5.10)	Compared to Meta-SNP > 0.7	3.55 (2.87–4.51)	Compared to Meta-SNP > 0.7	3.19 (3.02–3.37)
	0.4–0.5		55.1 (50.0–60.7)		25.3 (24.8–25.8)		14.4 (11.1–19.1)		10.4 (10.0–11.0)
	0–0.5		30.4 (27.1–34.1)		16.8 (16.4–17.3)		10.6 (8.32–14.2)		8.92 (8.42–9.45)
	0–0.4		34.3 (31.5–37.3)		18.2 (17.9–18.6)		19.4 (15.6–24.9)		18.1 (17.1–19.1)

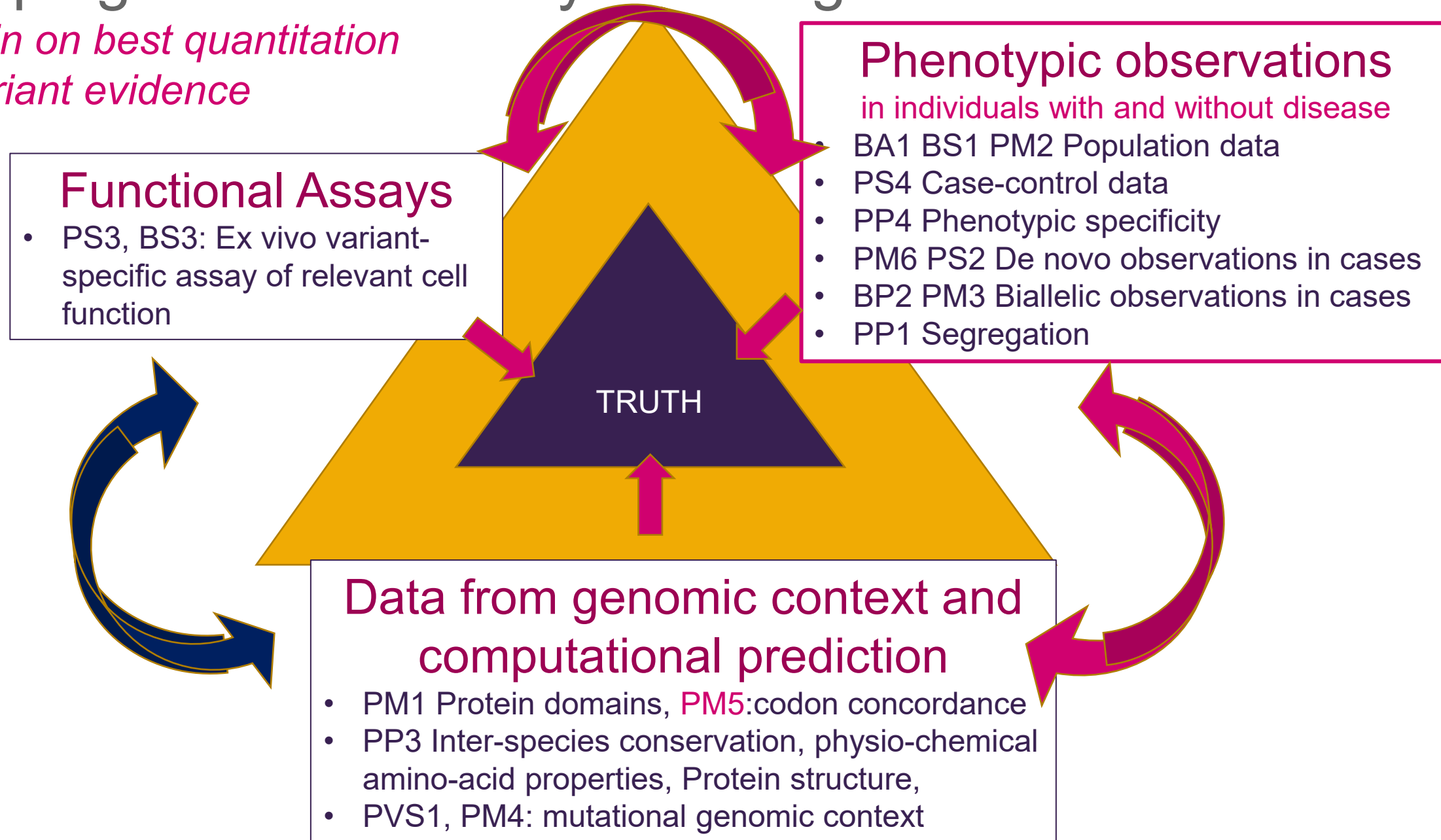
Quantifying and cross-calibrating evidence items

Functional assays ↔ PP3: In silico tools



Developing best community variant gold-truth

*Iterating in on best quantitation
of our variant evidence*



Quantifying and cross-calibrating evidence items

Functional assays ↔ PM5: With-codon concordance



Genetics
in
Medicine

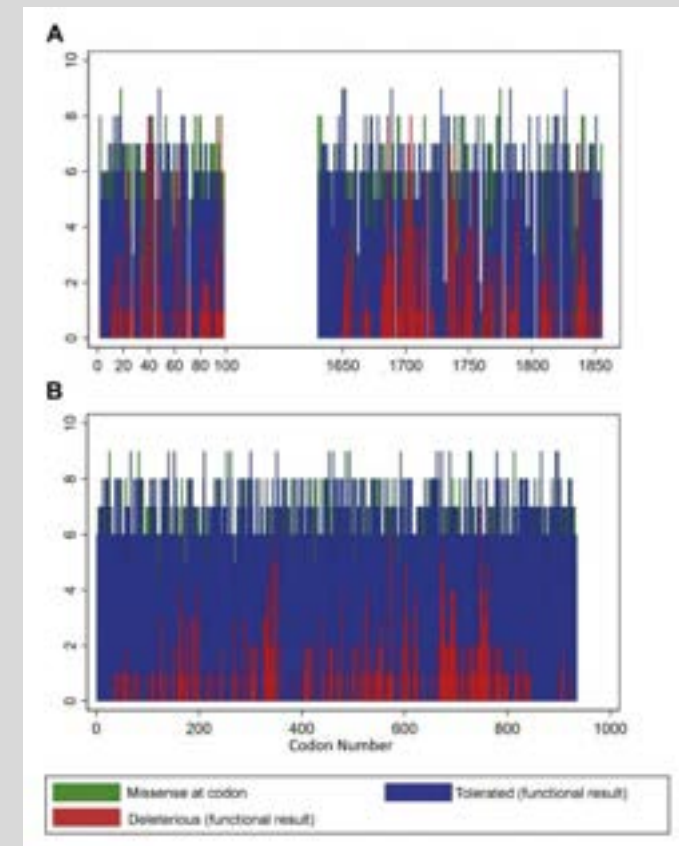
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ARTICLE

Quantifying prediction of pathogenicity for within-codon concordance (PM5) using 7541 functional classifications of *BRCA1* and *MSH2* missense variants

Table 2 Positive LR for nonoverlapping bands for PM5 for (1) *BRCA1* and *MSH2* combined, (2) *BRCA1*, and (3) *MSH2*

PM5-Definition-Band		Tool	BRCA1 + MSH2					BRCA1					MSH2				
			TP	FN	FP	TN	Positive LR	TP	FN	FP	TN	Positive LR	TP	FN	FP	TN	Positive LR
PM5_baseline_band: variants not attaining criteria for PM5_band_x or PM5_band_y																	
PM5_band_x) Exactly 1 deleterious colocated variant at codon; variant under examination has an equal or more damaging in silico score than colocated variant; comparison with baseline variant set	Revel	53	679	37	6688	13.1 (8.7-19.7)	31	310	11	1398	11.3 (5.8-22.0)	22	369	26	5290	11.5 (6.6-20.0)	
	Meta-SNP	64	656	23	6711	25.6 (16.1-40.9)	37	295	5	1404	28.9 (11.9-70.1)	27	361	18	5307	20.4 (11.4-36.4)	
	CADD	68	647	49	6663	13.0 (9.1-18.5)	41	285	21	1376	8.3 (5.0-13.7)	27	362	28	5287	13.2 (7.9-22.0)	
	Grantham Score	69	659	45	6690	14.1 (9.8-20.3)	38	304	18	1393	8.6 (5.0-14.7)	31	355	27	5297	15.8 (9.6-26.0)	
	aGVGD	69	660	35	6686	18.0 (12.1-26.8)	38	304	12	1399	12.7 (6.8-23.7)	31	356	23	5287	18.3 (10.9-31.0)	
	BLOSUM45	56	668	31	6700	16.7 (10.8-25.6)	33	303	8	1401	16.5 (7.8-34.7)	23	365	23	5299	13.7 (7.8-24.0)	
	BLOSUM62	56	662	32	6698	16.3 (10.6-24.9)	32	301	8	1401	16.1 (7.7-34.0)	24	361	24	5297	13.8 (8.0-23.9)	
	BLOSUM80	54	668	34	6697	14.7 (9.7-22.4)	28	307	9	1401	12.6 (6.1-26.0)	26	361	25	5296	14.3 (8.4-24.3)	
PM5_band_y) ≥2 deleterious colocated variants at codon; variant under examination has an equal or more damaging in silico score than ≥1 colocated variant; comparison with baseline variant set	Revel	69	679	15	6688	40.1 (23.3-69.2)	53	310	4	1398	45.8 (17.6-119.1)	16	369	11	5290	19.7 (9.3-41.5)	
	Meta-SNP	81	656	6	6711	114.1 (51.5-252.8)	62	295	4	1404	54.7 (21.1-141.3)	19	361	2	5307	108.7 (29.3-403.9)	
	CADD	86	647	28	6663	27.7 (18.2-42.0)	68	285	16	1376	16.3 (9.7-27.6)	18	362	12	5287	20.6 (10.1-41.9)	
	Grantham Score	86	647	28	6663	27.7 (18.2-42.0)	52	304	2	1393	82.1 (23.2-290.5)	21	355	3	5297	86.4 (28.0-266.0)	
	aGVGD	72	660	19	6686	34.0 (20.8-55.8)	52	304	2	1399	82.5 (23.3-291.7)	20	356	17	5287	16.5 (8.8-30.9)	
	BLOSUM45	77	668	9	6700	73.4 (37.6-143.3)	58	303	4	1401	50.5 (19.5-130.8)	19	365	5	5299	48.9 (19.1-125.1)	
	BLOSUM62	83	662	10	6698	71.5 (37.8-135.3)	61	301	4	1401	52.9 (20.5-136.9)	22	361	6	5297	47.8 (20.1-113.7)	
	BLOSUM80	79	668	9	6697	75.0 (38.5-146.4)	59	307	3	1401	65.1 (22.3-190.1)	20	361	6	5296	43.8 (18.2-105.1)	



Lucy Loong



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Complexities/challenges: MAVES-leveraging clinical impact

- Phenotypic evidence area critical
 - Improve classification evidence and generate truthsets
 - COMMON PHENOTYPES: Large **individual-level clinical variant data-sets** (with **denominators**) (PS4): ***quantify association signal***
 - RARE PHENOTYPES: **Accurate capture of phenotype** (severity and sub-phenotypes) (PP4): ***quantify specificity***
 - BENIGN TRUTHSETS ***low numbers for many genes***
- Cross-analyses **between evidence areas (Phenotypic, functional, predictive)** to finesse iteratively :
 - Evidence weighting of assay, tool, evidence items
 - Working variant truth-sets
 - Weed out outlying erroneous results/classifications
- Need:
 - universal infrastructure (ACMG framework, ClinGen SVI/SVC, FWG, VCEPs, ClinVar)
 - More data sharing

Themes

1. Clinical Genetics
2. Phenotypes and ascertainment
3. Variant-specific risks/genotype-phenotype
4. Evolution of Variant Interpretation Frameworks
5. MAVES, clinical variant interpretation and opportunities forward

Acknowledgements:

CanVIG-UK Steering Advisory Group (CStAG)

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Miranda Durkie (North East and Yorkshire GLH)
Rachel Robinson (North East and Yorkshire GLH)
George Burghel (North West GLH)
Jo Field (East GLH)
Alison Callaway (Central and South GLH)
Terri McVeigh (North Thames GLH)
Helen Hanson (South West GLH)
Bethan Frugtniet (South East GLH)
Trudi McDevitt (Ireland)
Sheila Palmer-Smith (Wales)
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Judith Pagan (Edinburgh)



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